GitHub □□ □□□□□□ & AI □□ □□ □□ □□ □□□□

> **□□:** *□□□ □□□ (□□□□□ □□, □□□□ □□□ □□)*
> **□□:** *GitHub Stars 10,000+ □□□□□ □□*
> **□□:** *Claude Code + GSTACK + Sequential Thinking + Plan □□ □□*
> **□□:** *4.0 (□□ □□ + □□□ □□□□□)*

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- GitHub □□ □□□□□ & AI □□ □□ □□ □□ □□□□
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  - Claude Code Plan □□
  - Sequential Thinking
  - GSTACK (Garry Tan's Skill Pack)
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- Part 1: □□ □□ □□□
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- □□□□ 4: □□□□□ □□ □□ □□□
- 4-1. CS □□ □□□
  - ossu/computer-science (178K+ Stars)
  - kamranahmedse/developer-roadmap (310K+ Stars)
  - jwasham/coding-interview-university (310K+ Stars)
- 4-2. □□ □□ - □□ □□□□
  - freeCodeCamp/freeCodeCamp (410K+ Stars)
  - TheOdinProject/curriculum (10K+ Stars)
  - codecrafters-io/build-your-own-x (330K+ Stars)
- 4-3. □□ □□
  - yangshun/tech-interview-handbook (122K+ Stars)
  - donnemartin/system-design-primer (290K+ Stars)
  - kdn251/interviews (63K+ Stars)
- 4-4. □□□ □□ & □□□
  - Python □□□ □□ □□
- Part 2: □□□□□ □ □□□ □□ □□□□□
- □□ □□
- 2-1. Spring Boot □□□
  - 2-1-1. spring-projects/spring-boot (76K+)

    - 2-1-2. eugenp/tutorials (37K+) — Baeldung □□ □□
    - 2-1-3. macrozheng/mall (83K+) — □□□□□ □□□
    - 2-1-4. iluwatar/java-design-patterns (90K+)
    - 2-1-5. gothinkster/realworld (80K+)
- 2-2. React □□□
    - 2-2-1. facebook/react (244K+)
    - 2-2-2. typescript-cheatsheets/react (45K+)
    - 2-2-3. alan2207/bulletproof-react (29K+)
- 2-3. Next.js □□□
    - 2-3-1. vercel/next.js (130K+)
    - 2-3-2. vercel/commerce (11K+) — □□□ □□□
- 2-4. □ □□ □□ □□□□□
- Part 3: □□□□ □ □□□ □□
- □□ □□
- 3-1. □□□ □□□□ □□□
    - 3-1-1. RealWorld (gothinkster/realworld)
    - 3-1-2. Full Stack FastAPI Template
    - 3-1-3. Redoc: API □□□ □□
- 3-2. □□□□□ & □□□□ □□
    - 3-2-1. First Contributions
    - 3-2-2. Awesome for Beginners
    - 3-2-3. GitHub Opensource Guide
- 3-3. □□□ & □□□
    - 3-3-1. Hackathon Starter
    - 3-3-2. AutoGPT
    - 3-3-3. LangChain
- Part 4: □□ □□ □□□□□ □□ □□□
- □□ □□
- 4-1. Python □□ □□□□□
    - 4-1-1. python-docx (Word □□)
    - 4-1-2. openpyxl (Excel □□)
    - 4-1-3. python-pptx (PowerPoint)
    - 4-1-4. PyMuPDF (PDF □□)
- 4-2. Java □□ □□□□□
    - 4-2-1. Apache POI (Excel/Word/PowerPoint)
    - 4-2-2. iText (PDF □□/□□)

- 4-3. JavaScript □□ □□□□□
  - 4-3-1. SheetJS (xlsx)
  - 4-3-2. PDFKit
  - 4-3-3. docx (Word □□)
- 4-4. □□ □□□ □□□□□
  - Claude Code□ □□ □□□□□ □□
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- Part 5: AI □□ □□ — Claude Code MCP & GSTACK
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- 5-1. □□ MCP □□
  - MCP □□
  - GitHub MCP - □□□□□ □□
  - Filesystem MCP - □□ □□ □□□
  - PostgreSQL MCP - □□□□□□ □□
  - Sequential Thinking MCP - □□□ □□
- 5-2. □□□ MCP □□
  - Playwright MCP - □□□□ □□□
  - Memory MCP - □□□□ □□ □□
  - Docker MCP - □□□□ □□
- 5-3. MCP □□ □□ □□
  - □□ □□□□□ (□□□ □□)
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- 5-4. GSTACK □□ □ □□
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  - 23□ GSTACK □□ □□
- 5-5. GSTACK □□□ □□ □□□
  - CEO □□: □□ □□
  - Engineer □□: □□ □□
  - QA □□: □□ □□
  - Release □□: □□ □□
- 5-6. GSTACK + MCP □□□ □□□□□
  - □□□ □□□□ □□□□□□
- 5-7. Plan □□ & Sequential Thinking □□□
  - Plan □□□?
  - Plan □□ □□□
  - Sequential Thinking □□□

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- 5-8. □□ □□ □□□□ □□
    - □□: "AI □□ □□ □" SaaS □□
- 5-9. GSTACK □ □ □□□ □□□□
    - □□□□ □□□
- Part 6: □□ □□ □ □□ □□
- □□ □□
- 6-1. □□ □□ □□
    - yangshun/tech-interview-handbook (122K+ Stars)
    - donnemartin/system-design-primer (290K+ Stars)
    - kdn251/interviews (63K+ Stars)
- 6-2. □□/□□□ □□□ □□□□ □□□□□
    - B-1. □□ IT □□ □□□□□
    - B-2. □□□ □□□□□ □□
    - B-3. □□ & □□□ (□□□ □□)
    - B-4. DevOps & CI/CD
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- □□ A: Claude Code Plan □□ □□ □□□
    - A-1. Plan □□□?
    - A-2. Plan □□ □□□ □□
    - A-3. Plan □□ vs Direct □□ □□ □□
    - A-4. Plan □□ □□□ □□□□□
    - A-5. Plan □□ (Ctrl+G)
    - A-6. Plan □□ □□□ □□□□
- □□ B: Sequential Thinking □□ □□□
    - B-1. Sequential Thinking MCP□?
    - B-2. Sequential Thinking □□
    - B-3. Sequential Thinking □□ □□
    - B-4. Sequential Thinking □□ □□
    - B-5. Sequential Thinking □□ □□
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```
src/*.md  →  md_merger.py  →  output/Guidebook.md
                                        ↓
                              ┌─────────┴─────────┐
                              ↓                   ↓
                        md_to_docx.py       md_to_pdf.py
                              ↓                   ↓
                   output/Guidebook.docx  output/Guidebook.pdf
```

- **python-docx** (Part 4-1)□ Word □□□ □□ □□
- **WeasyPrint** (Part 4-4 □□)□ PDF□ □□ □□
- **Pygments**□ □□ □□ □□□ □□

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Part 1	□□ □□ □□□	□□□□/□□□	CS □□, □□ □□, □□□ □□
Part 2	□□□□□/□□□ □□	□□~□□	Spring Boot, React, Next.js, Django □
Part 3	□□□□ □ □□□	□□	□□□ □□□□, □□□□□, □□□□ □□
Part 4	□□ □□ □□□□□	□□~□□	python-docx, openpyxl, WeasyPrint □
Part 5	AI □□ □□	□□ □□	Claude Code MCP, GSTACK □□□□□
Part 6	□□ □□/□□	□□□	□□ □□, □□/□□□ □□
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□□ □□□: Part 1 → Part 2 → Part 3 → Part 6

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AI □□□ □□: Part 5 → Part 4 → Part 2

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Claude Code Plan □□

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Sequential Thinking

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5. □□ □□□□□ □□

GSTACK (Garry Tan's Skill Pack)

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- `/office-hours` → □□ □□ □□ □□
- `/plan-ceo-review` → □□□ □□ □□
- `/plan-eng-review` → □□□□□ □□□□ □□
- `/qa` → □□□ □□ □□
- `/ship` → □□ □□□ □□

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```
# □ □□□□□□ □□ □□□□□□□!
git clone <repository-url>
cd PLAF

# □□□ □□
pip install -r scripts/requirements.txt

# □□□□ □□ (MD + DOCX + PDF)
python scripts/build_guidebook.py
```

□□□□ `output/` □□□□□ □□□□□.

Part 1: □□ □□ □□□

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4-1. CS □□ □□□□

ossu/computer-science (178K+ Stars)

□□: https://github.com/ossu/computer-science

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- □□ □□: Fundamentals → Core → Advanced → Specialization
- □□: □□□□ □□ □□ (□□□□ □□ □□)

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1. **Repository □□ □□**

```
# OSSU/CS □□□□ □□
git clone https://github.com/ossu/computer-science.git
cd computer-science

# README.md□□ □□□□□ □□ □□ □□
cat README.md | head -100
```

1. □□ □□ □□ (□□□□□ □□)

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1□□: □□	12□	Programming (Python), Math Foundations	□□
2□□: □□	24□	Data Structures, Algorithms, Database	□□
3□□: □□	16□	OS, Networks, Compilers	□□
4□□: □□□□	12□	ML, Security, Distributed Systems □ □□	□□□□

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Step 1: □□□□□ □□ (MIT CS50)
- CS50x Weeks 6-8□□ Python □□, □□, □□□/□□□□ □ □□□□ □□□□.
- □□ □□ □□□(check50)□□ □□ □□□□ □□ □ □□ □□□ □□□□□.

Step 2: □□□□ □□ (□□, □, □□□□□□)
- UC San Diego□ Data Structures □□□□ □□□ □□, OSSU □□□ □□ □□□ □□ □□ □□□□□.
- □□ □□□ □□□□□. □□□ □□□ □ Python `list`, `collections.deque` □ □□□□□.

Step 3: □□□□ □□ (□□/□□)
- Princeton Algorithms Part I□□ □□/□□/□ □□, □□ □□ □□ □□□□□.
- □□□ □□ □□ □ `coursework/` □□□ □□ □□□ □□□□ □□□□.

> *Claude Code □□ □*
> `> "OSSU CS □□□□□□ □□□□□□ □□ □□□ 3□ □□□ □□□ □□ □□□ □□"`
> `> "□ □□□ Core CS □□□ □□ □ □□(Python □□□ □)□ □□ □□□□ □□□□"`

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□ 1-4: Python □□ (MIT CS50)
- □□: CS50x Weeks 6-8 (Python)
- □□: □ 8-10□□
- □□□□: Hello World, □□□ □□□, To-do list □□□□

□ 5-8: □□□□ (UC San Diego DSA)
- □□: Algorithms and Data Structures
- □□: □ 12-15□□
- □□□□: Stack/Queue □□, LinkedList □□□

□ 9-12: □□□□ (Princeton Algorithms)
- □□: Algorithms Part I & II
- □□: □ 15-20□□
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4. □□□ □□□□: OSSU Discord □□□ □□□ □□ □□

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kamranahmedse/developer-roadmap (310K+ Stars)

□□: https://github.com/kamranahmedse/developer-roadmap

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```
roadmap/
├── Frontend Developer Roadmap
│   ├── 1단 HTML, CSS, JavaScript □
│   ├── 2단 □□(npm), □□(git)
│   ├── 3단 CSS □(Tailwind, Bootstrap)
│   ├── 4단 JavaScript □(DOM, AJAX)
│   └── 5단 □□(React, Vue, Angular)
├── Backend Developer Roadmap
│   ├── 1단 □□(Node.js, Python, Java)
│   ├── 2단 □□(Express, Django, Spring)
│   ├── 3단 □(SQL, NoSQL)
│   ├── 4단 API □(REST, GraphQL)
│   └── 5단 □& □(Docker, Kubernetes)
├── DevOps Roadmap
│   ├── Linux □
│   ├── □(Docker)
│   ├── □(Kubernetes)
│   ├── CI/CD (GitHub Actions, Jenkins)
│   └── □(Prometheus, ELK)
└── Full Stack Developer Roadmap
    └── Frontend + Backend + DevOps □
```

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```
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- □(0-6□월): Beginner roadmap □
- □(6-18□월): Intermediate roadmap □
- □(18□월): Advanced roadmap □
```

2□□: □□□□ □□□

```
Frontend □□□
Essential (□): HTML → CSS → JavaScript → React
Good to Have (□): TypeScript → Testing → Performance
Advanced (□): Custom Build Tools → Web Performance → PWA
```

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- □□□□□ □□□ 40-60□□□ □□□□. □ 20□□ □□ □□ □□□ 2-3□□ □□□□□.
- Frontend □□□ □: HTML → CSS → JavaScript → DOM → React → Hooks → State Management □□□ □ 3□.
- □□□□ □ □□□ "Personal Recommendation / Alternative Option"□□ □□□□ □□ □□□□□ □□ □ □ □□□□.

> *Claude Code □□ □*
> `> "developer-roadmap □□□ Frontend □□□□ □□ □ 20□□ □□ □□ 6□□ □□□□ □□□□"`

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- FreeCodeCamp □□□□
- Udemy □□
- MDN □□
- □□ □□

---

jwasham/coding-interview-university (310K+ Stars)

□□: https://github.com/jwasham/coding-interview-university

Google□ □□ □□□□ □□□□ □□ □□□ □□□ □□ □□□□□□.

📚 □□ □□

```
Interview Prep Curriculum
├── □(10□)
│   ├── Array
│   ├── LinkedList
│   ├── Stack
│   ├── Queue
│   ├── Hash Table
│   ├── Binary Search Tree
│   ├── Heap
│   └── Graph
├── □(20□)
│   ├── DFS (Depth First Search)
│   ├── BFS (Breadth First Search)
│   ├── Dynamic Programming
│   ├── Bit Manipulation
│   └── Combinatorics
├── □□(5□)
│   ├── Scalability
│   ├── Database Design
│   ├── Caching
│   └── Load Balancing
└── □□(3□)
    ├── STAR □
    ├── □□
    └── □□
```

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LeetCode Easy: Two Sum □□

```python
def two_sum(nums: list, target: int) -> list:
    """
    □□□ □□□□□□ □□ target□ □□ □ □□ □□□□ □□

    □: nums = [2, 7, 11, 15], target = 9
    □□: [0, 1] (2 + 7 = 9)
    """
    # □□ 1: Brute Force (O(n²))
    # for i in range(len(nums)):
    #     for j in range(i + 1, len(nums)):
    #         if nums[i] + nums[j] == target:
    #             return [i, j]

    # □□ 2: Hash Map (O(n)) - □□
    seen = {}
    for i, num in enumerate(nums):
        complement = target - num
        if complement in seen:
            return [seen[complement], i]
        seen[num] = i
    return []

# □□□
assert two_sum([2, 7, 11, 15], 9) == [0, 1]
assert two_sum([3, 2, 4], 6) == [1, 2]
print("✓ Two Sum □")
```

LeetCode Medium: Binary Search Tree Validate

```python
from typing import Optional

class TreeNode:
    def __init__(self, val=0, left=None, right=None):
        self.val = val
        self.left = left
        self.right = right

def is_valid_bst(root: Optional[TreeNode]) -> bool:
    """
    □□□ □□ □□□ □□□ □□ □□ □□□□ □□

    □□ □□ □□ □□:
    - □□ □□□□□ □□ □ < □□□
    - □□□ □□□□□ □□ □ > □□□
    """
    def validate(node, min_val=float('-inf'), max_val=float('inf')):
        if not node:
            return True

        if node.val <= min_val or node.val >= max_val:
            return False

        return (validate(node.left, min_val, node.val) and
                validate(node.right, node.val, max_val))

    return validate(root)

# □□□ □□□
#     2
#    / \
#   1   3
root = TreeNode(2)
root.left = TreeNode(1)
root.right = TreeNode(3)
assert is_valid_bst(root) == True
print("✓ Valid BST □")

# □□□ □□
#     5
#    / \
#   1   4
#      / \
#     3   6
bad_root = TreeNode(5)
bad_root.left = TreeNode(1)
bad_root.right = TreeNode(4)
bad_root.right.left = TreeNode(3)
bad_root.right.right = TreeNode(6)
assert is_valid_bst(bad_root) == False
print("✓ Invalid BST □")
```

LeetCode Hard: Median of Two Sorted Arrays

```python
def find_median_sorted_arrays(nums1: list, nums2: list) -> float:
    """
    □ □□□ □□□ □□□ □□ (O(log(min(m,n))) □□)
    """
    if len(nums1) > len(nums2):
        nums1, nums2 = nums2, nums1

    m, n = len(nums1), len(nums2)
    left, right = 0, m

    while left <= right:
        partition1 = (left + right) // 2
        partition2 = (m + n + 1) // 2 - partition1

        maxLeft1 = float('-inf') if partition1 == 0 else nums1[partition1 - 1]
        minRight1 = float('inf') if partition1 == m else nums1[partition1]

        maxLeft2 = float('-inf') if partition2 == 0 else nums2[partition2 - 1]
        minRight2 = float('inf') if partition2 == n else nums2[partition2]

        if maxLeft1 <= minRight1 and maxLeft2 <= minRight2:
            if (m + n) % 2 == 0:
                return (max(maxLeft1, maxLeft2) + min(minRight1, minRight2)) / 2
            else:
                return max(maxLeft1, maxLeft2)
        elif maxLeft1 > minRight2:
            right = partition1 - 1
        else:
            left = partition1 + 1

    return -1

# □□□
assert find_median_sorted_arrays([1, 3], [2]) == 2.0
assert find_median_sorted_arrays([1, 2], [3, 4]) == 2.5
print("✓ Median of Two Sorted Arrays □")
```

📅 8□ □□ □□□

```
1□ □□(Array, LinkedList, Stack, Queue)
   - □1□□□(LeetCode Easy)
   - □ 1-2□

2□ □□(Tree, Graph, Hash Table)
   - □1-2□□(Easy + Medium)
   - □ 2-3□

3□ □□□
   - □□ □□□
   - BFS, DFS □
   - □2□□(Medium)

4□ □□
   - DP □□
   - □DP □□
   - □1-2□□(Medium-Hard)

5□ □□□
   - Scalability □
   - Database □
   - □2-3□□□□

6□ □□□
   - STAR □□
   - □□
   - □□1□

7□ □□□□
   - □□□60□
   - □2-3□□□

8□ □□□□□
   - □1-2□□□
   - □□□□
```

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□□ □□ Checklist

```python
def technical_interview_checklist():
    checklist = {
        "□": [
            "□□□□",
            "□□□",
            "□□"
        ],
        "□": [
            "□□□",
            "□□□",
            "□□"
        ],
        "□": [
            "□□",
            "□□",
            "□□□□"
        ],
        "□": [
            "□□□",
            "□□□",
            "□□"
        ]
    }

    for category, items in checklist.items():
        print(f"\n[{category}]")
        for item in items:
            print(f"  □ {item}")

technical_interview_checklist()
```

4-2. □□ □□ - □□ □□□□

freeCodeCamp/freeCodeCamp (410K+ Stars)

□□: https://github.com/freeCodeCamp/freeCodeCamp

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```
freeCodeCamp Curriculum
├── Responsive Web Design (300□)
│   ├── HTML & CSS □
│   ├── Flexbox & Grid
│   └── 5□□(Portfolio, Survey, Landing Page □)
├── JavaScript Algorithms and Data Structures (300□)
│   ├── JavaScript □
│   ├── ES6+ □
│   ├── □& □
│   └── 5□□
├── Front End Development Libraries (300□)
│   ├── Bootstrap
│   ├── jQuery
│   ├── React
│   ├── Redux
│   └── 5□□
├── Data Visualization (300□)
│   ├── D3.js
│   ├── API & Fetch
│   └── 4□□
└── Back End Development and APIs (300□)
    ├── Node.js & Express
    ├── MongoDB
    ├── User Authentication
    └── 5□□
```

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Step 1: □□ □□ □□

```
# Node.js □□ □□
node --version
npm --version

# VS Code □□
# https://code.visualstudio.com/

# □□ □□ □□□□
# - Live Server
# - Thunder Client (API □□□)
# - Code Runner
# - Prettier (□□ □□□)
```

Step 2: □ □□ □□□□ **- Personal Portfolio**

- freeCodeCamp Responsive Web Design □□□□ HTML/CSS Flexbox·Grid□ □□ □ □□□□□ □□□□ □□ □□.
- □□□ □□: Navbar, Hero, About, Projects Grid, Contact. □□□(@media 768px □□)□ □□.
- Smooth scroll, sticky navbar □ JS □□□□□ □□ □□□□ □□□□ □□□□□.

> *Claude Code □□ □*
>
> `> "freeCodeCamp □□□□□ Responsive Web Design □□□□ 5□□ □□□□ □□□ □□□□□ □□□"`
>
> `> "□□ □□ index.html/style.css□ □□□□ □□□(a11y)·□□□ □□□□ □□□□ □□□"`

TheOdinProject/curriculum (10K+ Stars)

□□: https://github.com/TheOdinProject/curriculum

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```
The Odin Project Path
├── Foundations (10-20□)
│   ├── Installing Tools
│   ├── How This Course Will Work
│   ├── How the Web Works
│   ├── JavaScript Basics
│   ├── HTML Basics
│   ├── CSS Basics
│   ├── Flexbox
│   ├── Grid
│   └── 10□□
├── Intermediate HTML and CSS (4-8□)
│   ├── Intermediate HTML Concepts
│   ├── Intermediate CSS Concepts
│   ├── 2□□
├── JavaScript (12-20□)
│   ├── Fundamentals
│   ├── DOM Manipulation
│   ├── OOP
│   ├── Async JavaScript
│   └── 10□□
├── Getting Hired (□□□)
│   ├── Portfolio Project
│   ├── Preparing for Interviews
│   ├── Getting Hired
└── Paths (□□)
    ├── Foundations → React (Front-end)
    │   ├── React Basics
    │   ├── Advanced React
    │   ├── 5□□
    │   └── Capstone Project
    └── Foundations → Rails (Full Stack)
        ├── Ruby Basics
        ├── Ruby on Rails
        ├── Databases
        ├── 5□□
        └── Capstone Project
```

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1. □□□□□ □□
- The Odin Project Foundations □□□ □ JavaScript □□□□. 3□ □□ + □□ □□ + □□ □□□□□ □□□□□.
- □□ □□□: `Math.random()`, □□□, DOM □□□ □□□, □□ □□.

2. □□□ □□□□ (□□)
- ES6 Class □□, □□ □□(`currentOperand`, `previousOperand`), □□□ □□(`[data-number]`) □□□ □□ □□.
- □□□ □□□ □□ "□□□ 2□ □□ □□", "0□□ □□□ □□" □□ □□ □□□□ □□ □□□□ □□□ □□□□.

> *Claude Code □□ □*
> `> "The Odin Project□ Calculator □□□□ □□□□□ □□ □□ □□ □□□ □□□ □□□□"`

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codecrafters-io/build-your-own-x (330K+ Stars)

□□: https://github.com/codecrafters-io/build-your-own-x

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```
Build Your Own X
├── Build Your Own Git (Rust)
│   ├── 1. Read Objects
│   ├── 2. Reading Commits
│   ├── 3. Read Tree Objects
│   ├── 4. Writing Objects
│   └── 5. Creating Commits
├── Build Your Own Docker (Go)
│   ├── 1. Container Basics
│   ├── 2. Namespaces
│   ├── 3. Cgroups
│   └── 4. Docker CLI
├── Build Your Own Redis (Python/Ruby/Go)
│   ├── 1. TCP Server
│   ├── 2. RESP Protocol
│   ├── 3. Expiration
│   └── 4. Replication
└── Build Your Own Language (Go/Python)
    ├── 1. Lexer
    ├── 2. Parser
    ├── 3. Evaluator
    └── 4. Advanced Features
```

💾 □□: Redis □□□ (Python)

- □□ □ `Build Your Own Redis` □□□□□ □□□ □□ □□□ □□□□□.
- □□ □□ □□: TCP □□ □□ → RESP □□□□ □□ → SET/GET/DEL □□□ → □□(EX □□) → □□.
- Python `socket`, `threading`, `datetime` □ □ □□□□□ □□ □□□ □□□□□.

> *Claude Code □□ □*
> `> "build-your-own-x □□□□ Build Your Own Redis □□□ □□, □□□□□ 1□□□ □□□□ □ □□ □□ □□ □□□□ □□□"`

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```
□(1□)
- Git □□□
- □TCP □□
- □□□

□(2-3□)
- Redis □□□
- Docker □□
- □□

□(3-6□)
- □□□
- □□□
- □□
```

4-3. □□ □□

yangshun/tech-interview-handbook (122K+ Stars)

□□: https://github.com/yangshun/tech-interview-handbook

□□□ □□□ □□ □□ □□□. □□, □□, □□ □□□□ □□ □□□□.

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1. □□ □□ (Coding Interview)

```python
# □□ □ □□□□□
interview_checklist = {
    "□□": [
        "□□□□",
        "□□□",
        "□LinkedIn □",
        "□□□3-5□□"
    ],
    "□□": [
        "□□□",
        "□□□",
        "□□□",
        "□□",
        "□□"
    ],
    "□□": [
        "□□",
        "□□□",
        "□□□",
        "□□"
    ]
}

for phase, items in interview_checklist.items():
    print(f"\n📌 {phase}")
    for item in items:
        print(f"  □ {item}")
```

2. □□□ □□□ □□

```
□□

1. □□□(5분)
   - □□(FRs)
   - □□(NFRs)
   - Scale, QPS, Data volume

2. API □(5분)
   - Endpoints □
   - Request/Response □
   - Rate limiting

3. □□□(5분)
   - Database schema
   - Indexing
   - Sharding strategy

4. □□□(10분)
   - Components
   - Data flow
   - Caching layer
   - Message queue

5. □□(10분)
   - Core components □
   - Bottleneck □
   - Optimization

6. □(5분)
   - Monitoring
   - Logging
   - Future improvements
```

□□ □□□ □□□ □: URL Shortener

```
1□ □
- □ □URL → □URL □ □
- □
  * □□ 500 writes/sec
  * □□ 100,000 reads/sec
  * □ 99.9%
  * Latency: < 200ms

2□ □□
- 5□□ 500 * 86400 * 365 * 5 = 78.8 billion URLs
- □□ 78.8B * 100 bytes = 7.88 TB

3□ API □
POST /api/shorten
{
    "long_url": "https://example.com/very/long/url"
}
□
{
    "short_url": "https://short.url/abc123"
}

GET /api/{short_code}
□ 301 Redirect to original URL

4□ □□
Table: URLMapping
- id (PK): bigint
- short_code: varchar(10) UNIQUE
- long_url: varchar(2048)
- created_at: timestamp
- expiration: timestamp

5□ □□□
Client → Load Balancer → API Servers
                            ↓
                        Cache (Redis)
                            ↓
                      Database (MySQL)
                            ↓
                      Message Queue (Kafka)
                            ↓
                      Analytics Service

6□ □□
- Short code □ Base62 encoding (0-9, a-z, A-Z)
- Collision □ Retry with increment
- Caching □ LRU cache in Redis
- Load □ Consistent hashing
```

---

donnemartin/system-design-primer (290K+ Stars)

□□: https://github.com/donnemartin/system-design-primer

□□ □□□ □□□ □□□ □□ □□ □□□□□□.

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1. Scalability (□□□)

```
Vertical Scaling (□□)
- CPU □
- □□
- 💰 □□□□

Horizontal Scaling (□□)
- □□□□
- Load balancer□□□
- 💰 □□□□□□

💰 100 QPS → 1000 QPS□□
Before (□□):
  [Application] → [Database] → Bottleneck!

After (□□):
  Load Balancer
      ↓
  App1  App2  App3
      ↓   ↓   ↓
    Cache (Redis)
      ↓
    Database (Master-Slave)
      ↓
  Slave1  Slave2
```

2. □□□□□□ □□

```
SQL vs NoSQL □

SQL (□□)
□
- ACID □
- □□□
- □□
□
- □□□
- □□

□ MySQL, PostgreSQL

NoSQL (□□)
□
- □□□
- □□□
- □□
□
- ACID □□□
- □□□
- Consistency □

□ MongoDB, Cassandra, DynamoDB
```

3. □□ □□
- LRU(Least Recently Used), LFU(Least Frequently Used), Write-Through/Write-Back □ □□ □ □□□ □□□□□.
- Python□□□ `collections.OrderedDict` □□ `functools.lru_cache` □□□□□□ □□ □□□ □ □□□□.
- system-design-primer□ □□ □□□□ □□□(Memcached, Redis, CDN)□□ □ □□□ □□□ □□□□ □□□ □□ □□□ □□.

---

kdn251/interviews (63K+ Stars)

□□: https://github.com/kdn251/interviews

Java □□ □□ □□□ □□□□ □ □□□□ □□.

🎯 Java □□ □□ □□

- □ □□□ LeetCode/HackerRank □□ □□□ Java □□□ □□□□□(Array, String, Tree, DP □)□ □□□□□□□□.
- □□□: □□□ □□ □□ □□□ □, □ □□□ □□□ □□/□□ □□□□ □□□□ □□□ □□□□□.
- Two Sum, Valid Palindrome, Reverse Integer □□ Easy □□□ □□□□ □□ Medium/Hard□ □□□□ □.

> *Claude Code □□ □*
> `> "kdn251/interviews □□□ Array □□□□ □□□ □□ □□ Top 10□ □□□"`

4-4. □□□ □□ & □□□

Python □□□ □□ □□

□□ □ □□ □□

```
# □□ □□ □□
python -m venv data-env
source data-env/bin/activate  # Linux/Mac
data-env\Scripts\activate  # Windows

# □□ □□□□□□ □□
pip install jupyter numpy pandas matplotlib seaborn scikit-learn streamlit
```

📊 Pandas□ □□□ □□□ □□

□□ API 5□□□ □□ 80%□ □□□ □□□□□.

```
import pandas as pd

df = pd.read_csv('data.csv')
df.head(); df.info(); df.describe()           # □□
df.fillna(df.mean(numeric_only=True))         # □□ □□
df.drop_duplicates()                          # □□ □□
df.groupby('category')['sales'].sum()         # □□ □□
df.corr(numeric_only=True)                    # □□□□
```

□□□□ `matplotlib` / `seaborn` □□ □□(hist, boxplot, scatter, heatmap)□ □□□□,
□□□□□ `scikit-learn` □ `train_test_split` → `StandardScaler` → `RandomForestClassifier`
3□□ □□□□□□ □□ □□□□.

📊 Streamlit□□ □□□□ □□□

- Streamlit□ Python □□□□ □ □□ □ □□□□□ □□□ □□□□□. HTML/CSS/JS □□□ □□□.
- □□ API: `st.title`, `st.sidebar`, `st.metric`, `st.line_chart`, `st.bar_chart`, `st.dataframe`, `@st.cache_data`.
- □□: `streamlit run dashboard.py` □ □□ □□□ □□□.

> *Claude Code □□ □*
> `> "□ CSV □□(□□: date, category, sales)□ Streamlit □□□□ □□□ □□□□. KPI 4□ + □□ □ □□ + □□□□ □□ □□□□."`

```
# Streamlit □□
streamlit run dashboard.py
```

Part 2: □□□□□ □ □□□ □□ □□□□□

> □□ □□□ □□□ □□ □□□□□□□ □□□ □□ □□□□□□□□.
> *Spring Boot, React, Next.js □□ □□□□□ □□□ □ □□□□.*

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- □□□ □□□□□ (Spring Boot) □□□ □□
- □□□□□ □□□□□ (React, Next.js) □□ □□ □□
- □ □□□□□□□ "□□□ □□ □□□ □□□" □□
- Claude Code□ □□ □□□□ □□□□□□ □□□□ □□ □□

□ □□□ 10,000+ Stars□ □□□ □□ □ □□ □□□□□ □□□□□□ "□□□ □□□□, □□□ □□, □□□ □□□" □□□□ □□□□□. □□ □□□ □□ □□ □□ □□ □□ · □□ □□ □□ · **Claude Code** □□ □□□ □□□□□.

2-1. Spring Boot □□□

2-1-1. spring-projects/spring-boot (76K+)

□□□□□ □□

Spring Boot□ Spring Framework □□□ □□ □□□□□□ □□(RAD) □□□□□□□□. □□ □□(Auto-Configuration)□ □□ □□□ □□□ □□□ □□□□□ □□□□ □□ □□□ □□□□□□. Java □□□ □□□ □□□ □□□□ □□□□□□□ □□□□□ □□□ □□□.

- Stars: 76,000+
- Language: Java
- License: Apache 2.0
- □□ □□□: https://spring.io/projects/spring-boot

□□ □ □□ □□

```
# JDK 17+ □□ □□
java -version

# □□□□□ □□ (□□□ □□□ shallow clone □□)
git clone --depth 1 https://github.com/spring-projects/spring-boot.git
cd spring-boot
```

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- `spring-boot-project/spring-boot` — □□ □□, `@SpringBootApplication` □□
- `spring-boot-project/spring-boot-autoconfigure` — □□ □□ □□ □□
- `spring-boot-project/spring-boot-starters` — □□□ □□□ □□
- `spring-boot-project/spring-boot-docs` — □□ □□□□ □□ □□
- `spring-boot-tests/spring-boot-smoke-tests` — □□ □□ □ □□(□□□ □□)

□□ □□: □ □□ □□ start.spring.io □□

□□ □□ □□□□□ https://start.spring.io/ □□ □□□□, spring-boot □□□ □□ □□□ □□□ □□ □□□□ □□ □□□□.

```
# Spring Initializr CLI (□□)
curl https://start.spring.io/starter.zip \
  -d dependencies=web,data-jpa,h2 \
  -d name=demo -o demo.zip
```

Claude Code □□ □

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> *"spring-boot-project/spring-boot-autoconfigure □□□□□□ DataSourceAutoConfiguration□ □□□ □□□□ □□□□□ □□□ □."*
>
> *"@SpringBootApplication □ □□□□□ □□ □□□□□□ □□□□□ □□□□ □□ □."*
>
> *"spring-boot-smoke-tests □ REST API □□ □□ □□□ □□ □□ □□□ □□□ □."*

2-1-2. eugenp/tutorials (37K+) — Baeldung □□ □□

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Baeldung □□□□ □□ □□ □□□ □□□□ □□□ Java/Spring □□ □□□□□□□. Spring Security, JPA, Kafka, Reactive, JWT □ □□ □□ □□□ □□□□ □□ □□□ □□□□□.

- Stars: 37,000+
- □□: Maven □□ □□ (□□ □)
- □□ □□□: https://www.baeldung.com/

□□ □ □□□ □□□ □□

```
git clone --depth 1 https://github.com/eugenp/tutorials.git
cd tutorials

# □□ □□□ 30□ □□ □□□□. □□□ □□□ □□□□□□.
cd spring-security-modules/spring-security-web-login
mvn spring-boot:run
```

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- `spring-boot-modules/` — Spring Boot □□
- `spring-security-modules/` — □□/□□ (JWT, OAuth2 □□)
- `persistence-modules/spring-data-jpa-*` — JPA, Query □□□
- `testing-modules/` — JUnit 5, Mockito, Testcontainers

Claude Code □□ □□

> *"spring-security-modules □□□□□□ JWT □□ □□□□ □□□ □□ □□□ □□□ □□ README□ □□ □□ □□□ □□ □ □."*
>
> *"persistence-modules □□□ Spring Data JPA* `@Query` *□□□□□ □□ □□□ 3□ □□ □□□ □."*

2-1-3. macrozheng/mall (83K+) — □□□□□ □□□

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□□ □□ □□□ □□□ □□□ □□□ □□□□□. Spring Boot + MyBatis + Elasticsearch + Redis + RabbitMQ + MongoDB □□□ □□ □□□□ □□□□□ □□ □□□ □□ □□□□□□□.

- Stars: 83,000+
- □□: mall-admin(□□□), mall-portal(□□□), mall-search(□□), mall-demo

□□ □ □□ □□

```
git clone --depth 1 https://github.com/macrozheng/mall.git
cd mall

# □□ □□□: MySQL 5.7+, Redis, (□□)Elasticsearch/RabbitMQ/MongoDB
# Docker Compose □□□ □□ □□
# document/docker/ □□□□□ □□ □□ □□
```

□□□□ □□□□ □□□ □

- `mall-admin/` — □□□ □□□□ API
- `mall-portal/` — □□□ □□□ API (□□, □□□□, □□)
- `mall-search/` — Elasticsearch □□ □□ □□
- `document/` — □□□□ □□□□□ · DB □□□ · API □□ (□□□ □□ □□)

Claude Code □□ □□□□

> *"mall-portal □□□□ □□ □□(Order)□□ □□ □□□□ □□□ Service □□ □□□ □□□ □."*
>
> *"mall □□□□□ Redis □□ □□□ □□ Service □□□□ □□□□□ □□□□ □."*
>
> *"document □□□□□ DB □□□□ □□ □□-□□□□-SKU □□□ ER □□□□□□□ □□□ □."*

2-1-4. iluwatar/java-design-patterns (90K+)

□□□□□ □□

GoF □□□ □□ + □□□□□□ □□□ Java□ □□□ □□ □□ □□ □□□□□. □ □□□□ □□ Maven □□, □□□ □□□□□, README□ □□□□ □□ □ □□□ **30**□□□ □□□ □ □□□□.

- Stars: 90,000+
- □□ □: 150+ (Creational, Structural, Behavioral, Concurrency, Architectural □)

□□ □ □□ □□ □□

```
git clone --depth 1 https://github.com/iluwatar/java-design-patterns.git
cd java-design-patterns

# □□□ □□ □□□□□ □□ □ □□
cd singleton
mvn compile exec:java
```

□□□□ □□ □□ □ □□

- `singleton/`, `factory/`, `builder/` — Creational □□
- `strategy/`, `observer/`, `command/` — Behavioral □□
- `adapter/`, `decorator/`, `facade/` — Structural □□
- `repository/`, `dao/` — □□□ □□□□ □□ □□

Claude Code □□ □□

> "□ □□□ *strategy* □□ *README*□ *App.java*□ □□, □□ □□ □□□(□□/□□□□□/□□)□ □□□ □□□□ □□□ □□□ □."
> "*decorator* □□□ *chain-of-responsibility* □□□ □□□ □ □□ □□□ □□□ □□□ □."

2-1-5. gothinkster/realworld (80K+)

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"□□ □□□□□□ □□□ □□□ □□□ □(Conduit)□ □□□□"□ □□□ □□ □□□□□□□□□. □□□□□ · □□□ **100+** □□□□ □□ □□□ API □□□ □□□□, □□ □□□ □□ □□ □□□□ □□ □□□ □ □□□□.

- Stars: 80,000+
- API □□: https://docs.realworld.show/

□□ □□

```
# □□ □□ □□ (□□□ □□□ □□□ □□)
git clone --depth 1 https://github.com/gothinkster/realworld.git

# □□□ □□□□ □□ □□□ □□□□□. □: Spring Boot □□
git clone https://github.com/gothinkster/spring-boot-realworld-example-app.git
```

□□ □□□

- □□ □□□ React / Vue / Angular / Svelte□ □□
- □□ API□ Spring Boot / Express / Django / Rails / NestJS□ □□
- □□ □□□ □□□□ □□ □□ □□□ □□□□ □□□ □□ □□

Claude Code □□ □□

> "*Spring Boot* □□□(*gothinkster/spring-boot-realworld-example-app*)□ *Express* □□□ (*gothinkster/node-express-realworld-example-app*)□□ *Article* □□ *API* □□□□ □□□ □□□ □."

2-2. React □□□

2-2-1. facebook/react (244K+)

□□□□□□ □□

React □□□ □□ □□ □□□□□□□□□. □□ □□□□□□ □ □□□□□ □□□□ □□, **Vite / Next.js / Create React App** □ □ □□ □□□□□□. □ □□□ **React** □□ □□□ □□□□ □□ □ □□□□□.

- Stars: 244,000+
- License: MIT
- □□ □□: https://react.dev/

□□ □□□□ □□ (□□ □□)

```
# Vite □□ (□□ □□)
npm create vite@latest my-app -- --template react-ts
cd my-app
npm install
npm run dev
```

□□□□ □□□ □

- `packages/react` — □□ □□□□□, Hooks □□
- `packages/react-dom` — DOM □□□
- `packages/react-reconciler` — Fiber □□□□
- `packages/scheduler` — □□□□ □□ □□□□

Claude Code □□ □□

> *"packages/react/src/ReactHooks.js □ □□ useState□ □□□□□ □□ □□□ □□□□□ □□□ □."*
>
> *"Fiber □□□□□ □ □□□□□ packages/react-reconciler README□ □□ □□□ □□□ □□□ □."*

2-2-2. typescript-cheatsheets/react (45K+)

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React + TypeScript □□□□ □□ □□ □□ □□□ □□□□□ □□□ □□□□ □□□□□. □□ □□ □□□□□□□□ □□ □□□ □□□□□. □□ □□ □□ □□□ □□□ □ □□ □□ □□□ □□□□.

- Stars: 45,000+

- □□: basic / advanced / migrating / hoc □□□□

□□ □□

```
# □□ □□ □□ □□□□ □□ □□ □□
# https://react-typescript-cheatsheet.netlify.app/

git clone --depth 1 https://github.com/typescript-cheatsheets/react.git
```

□□ □□□□ □ □□

- `README.md` — □□ □□□□
- `docs/basic/` — Props / State / Event / Forms □□ □□
- `docs/advanced/` — Generic □□□□, Polymorphic □□□□
- `docs/hoc/` — HOC□ Render Prop□ □□□

Claude Code □□ □□

> "□ □□ *docs/basic* □ □□□□ *Input* □□□□□ *onChange* □□□ □□□ □□□□ □□ *3*□□□ □□□ □."
> "*advanced* □□□□ '*Generic Components*' □□□ □□ □ □□□□□ *List* □□□□□ □□□ □□□□ □□□ □."

2-2-3. alan2207/bulletproof-react (29K+)

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"□□ □□□ **React** □□□□ □□"□ □□ □□□□ □□ □□□□ □□□□□. feature □□ □□ □□, □□ □□ □□, □□□ □□, □□ □□□□ □□ □□□ □□□□□ □□□ □□□ □□□□□□.

- Stars: 29,000+
- □□: React + TypeScript + Vite + Zustand + React Query + MSW

□□ □ □□

```
git clone --depth 1 https://github.com/alan2207/bulletproof-react.git
cd bulletproof-react/apps/react-vite
npm install
npm run dev
```

□□□□ □□□ □□ (□□)

- `docs/` — □□□□ □□ □□ (□□ □□)
- `src/features/` — □□□ □□ (auth, comments, discussions, users □)

- `src/components/` — □□ UI □□□□
- `src/lib/` — □□ □□□□□ □□ (axios, react-query □□)
- `src/testing/` — □□□ □□ + MSW □□□

Claude Code □□ □□□□

> *"docs □□□□□ project-structure.md □ application-overview.md □ □□ □ □□□□□ □□ □□ □□□ □ □□□□□ □□□ □ □□ □□□□□□ □□□ □."*
>
> *"src/features/discussions □□□□□ api, components, routes, types □ □□□ □□□□□□□ □□□ □."*

2-3. Next.js □□□

2-3-1. vercel/next.js (130K+)

□□□□□ □□

React □□ □□□ □□□□□. App Router · Server Components · Server Actions · □□□ □□□ · □□ □□ □□□ □ □□ □ □□□ □□□ □□ □□□ □□□□□. □□ □□□□□ `create-next-app` □□ □□□□, □ □□□ □□ □□ □□□ □□ □□□ □□□□ □.

- Stars: 130,000+
- □□ □□: https://nextjs.org/docs

□□ □□□□ □□

```
npx create-next-app@latest my-app --typescript --eslint --tailwind --app
cd my-app
npm run dev
# http://localhost:3000
```

Next.js □□□□ □□□ □

- `examples/` — □□ □□. 100+ □□ □□(□□, DB, □□, i18n □)
- `packages/next` — □□□□□ □□
- `docs/` — □□ □□ □□
- `test/` — E2E □□□ (□□ □□□□□ □□)

□□ □□ □□□□ (examples/ □□)

- `with-supabase` — □□ + DB □□□

- `with-stripe-typescript` — □□ □□
- `blog-starter` — □□□□ □□□
- `with-next-auth` — □□ □□□
- `api-routes-rest` — REST API □□

Claude Code □□ □□

> *"vercel/next.js □□□ examples/blog-starter □□□ □□□□, □ □□□□ □□□□□ □□□ □ □□□□ □ □□□ □□ □."*
>
> *"examples/with-next-auth □ □□ Google OAuth □□□□ □□□□ □□ □□ □□□ □□□ □."*
>
> *"App Router□ layout.tsx □ Pages Router□ _app.tsx □□□ □□ docs □□□□□□ □□□ □□ □□□ □."*

2-3-2. vercel/commerce (11K+) — □□□ □□□

□□□□□ □□

Next.js + Shopify □□ □□□□□ □□□ □□□□□. Server Components, Server Actions, □□ □□ □ □□ Next.js □□□ □□ □□□ □□ □□ **App Router** □□□□ □□□ □□□ □□ □□□□□.

□□ □□

```
git clone --depth 1 https://github.com/vercel/commerce.git
cd commerce
# README□ Shopify □□ □□□ □□□ □□ □□
```

□□□ □

- `app/` — App Router □□, □□□ □□, □□/□□□□ □□□
- `components/` — □□ UI + cart □□□□
- `lib/shopify/` — □□ API □□ □□ (Server-only □□)

Claude Code □□ □□

> *"vercel/commerce □ app/product/[handle]/page.tsx □ □□□ Shopify □□□□ Server Component □□ □□□□□ □□□ □."*
>
> *"□ □□□ revalidateTag □□ □□□ □ □□□ □□□□□ □□□ □□□□ □□□ □."*

2-4. □ □□ □□ □□□□□

- [ ] Spring Boot: `start.spring.io` □ □□ □□□□□ □□ □□□ □□□□
- [ ] eugenp/tutorials □□ □□ □□ 1□ □□□ □□ README□ □□□□
- [ ] iluwatar/java-design-patterns □□ 3□ □□□ □□ □□□ □□
- [ ] Vite + React + TypeScript □□□□□ □□□ Hooks □□□ □□□□
- [ ] bulletproof-react □ `docs/` □□□□□ □□ □□□
- [ ] `create-next-app` □□ App Router □□□□□ □□□ 1□ □□□□ □□□□
- [ ] vercel/next.js □ `examples/` □□ □□ □□ □□ 1□□ □□□ □□□□
- [ ] □ □□□ □□ Claude Code □□ □□ 1□ □□□ □□ □□□ □□□

> □□ □□: ***Part 3 —*** □□□□ □ □□□ □□□□□
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Part 3: □□□□ □ □□□□ □□

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3-1. □□□□ □□□□□ □□□□

3-1-1. RealWorld (gothinkster/realworld)

GitHub Stars: 80K+
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RealWorld□ □□□□ □□□□□□□(Spring Boot, React, Django, Node.js □)□ □□□□ □□□□□□□□□□ □□□□ □□ □□□□□□□. □□□□ □ □□□□□ "Conduit"□ □□□□□□, □□□□ □□, □□□□ CRUD, □□□□ □□, □□□□□□ □□ □□□□□□.

□□ □□

```
# RealWorld □□ □□
git clone https://github.com/gothinkster/realworld.git
cd realworld

# □□□ □□□ □□ □□ (□: Spring Boot)
cd implementations/spring-boot-gradle-jpa

# □□□ □□
./gradlew build

# □□□□□□ □□ (application.yml)
# spring.datasource.url: jdbc:h2:mem:testdb
# spring.jpa.hibernate.ddl-auto: create-drop
```

□□ □□ □□

User □□□ (Spring Boot)

```
@Entity
@Getter
@Setter
@NoArgsConstructor
public class User {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(unique = true)
    private String username;

    @Column(unique = true)
    private String email;

    private String password;
    private String bio;
    private String image;

    @OneToMany(mappedBy = "author")
    private List<Article> articles = new ArrayList<>();

    @ManyToMany
    @JoinTable(
        name = "user_followers",
        joinColumns = @JoinColumn(name = "user_id"),
        inverseJoinColumns = @JoinColumn(name = "follower_id")
    )
    private Set<User> followers = new HashSet<>();
}
```

Article □□□

```java
@Entity
@Getter
@Setter
public class Article {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String slug;
    private String title;
    private String description;
    private String body;

    @CreationTimestamp
    private LocalDateTime createdAt;

    @UpdateTimestamp
    private LocalDateTime updatedAt;

    @ManyToOne
    private User author;

    @ManyToMany
    @JoinTable(name = "article_favorites")
    private Set<User> favoritedBy = new HashSet<>();

    @ElementCollection
    private List<String> tagList = new ArrayList<>();
}
```

Article Service

```java
@Service
@RequiredArgsConstructor
public class ArticleService {
    private final ArticleRepository articleRepository;
    private final UserRepository userRepository;

    public ArticleResponse createArticle(String username, CreateArticleRequest request) {
        User author = userRepository.findByUsername(username)
            .orElseThrow(() -> new UserNotFoundException("User not found"));

        Article article = new Article();
        article.setTitle(request.getTitle());
        article.setDescription(request.getDescription());
        article.setBody(request.getBody());
        article.setAuthor(author);
        article.setSlug(generateSlug(request.getTitle()));
        article.setTagList(request.getTagList());

        Article saved = articleRepository.save(article);
        return ArticleResponse.fromArticle(saved);
    }

    public Page<ArticleResponse> listArticles(Pageable pageable, String author, String tag) {
        Specification<Article> spec = (root, query, cb) -> cb.conjunction();

        if (author != null) {
            spec = spec.and((root, query, cb) ->
                cb.equal(root.get("author").get("username"), author)
            );
        }

        if (tag != null) {
            spec = spec.and((root, query, cb) ->
                cb.isMember(tag, root.get("tagList"))
            );
        }

        return articleRepository.findAll(spec, pageable)
            .map(ArticleResponse::fromArticle);
    }
}
```

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□□□□ **1:** □□ □□□□ □□ □ □□
- Spring Boot □□□ + React □□□□□ □□□□ □□
- □□ □□, □□□ □□, □□ □□ □□
- □□□□□□ GitHub□ □□

□□□□ **2:** □□□□□□ □□□ □□
- H2 → PostgreSQL □□□□□□
- N+1 □□ □□ (Eager Loading vs Lazy Loading)
- □□ □□□ □□□ □□ □□

□□□□ **3:** □□ □□□□□ □□
- Docker □□□□□
- GitHub Actions CI/CD
- □□□□ □□ (AWS EC2, Heroku)

Claude Code □□ □

```
/think → RealWorld □□
- Entity □□□
- API □□□

/review → □□□
- JPA □□□
- □□□□

/test → □□□□
- ArticleService □□
- □□□
```

3-1-2. Full Stack FastAPI Template

GitHub Stars: 28K+
□□□□ □□: FastAPI + React + PostgreSQL + Docker □□□ □□□ □□□□□□□
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tiangolo/full-stack-fastapi-template□ □□□□ □□□ □□□ □□□□□□. Docker Compose□ □□ □□□□ □ □□ □ □□□, JWT □□, SQLAlchemy ORM, React □□ □□□ □□□□.

□□ □□

```
# □□ □□
git clone https://github.com/tiangolo/full-stack-fastapi-template.git
cd full-stack-fastapi-template

# Docker Compose□ □□ □□ □□
docker-compose up -d

# □□□□□□ □□
# □□□□□: http://localhost:3000
# API □□: http://localhost:8000/docs
```

□□ □□ □□

FastAPI User □□

```python
from fastapi import FastAPI, Depends
from sqlalchemy.orm import Session
from pydantic import BaseModel

app = FastAPI()

class UserBase(BaseModel):
    email: str
    full_name: str = None

class UserCreate(UserBase):
    password: str

class User(UserBase):
    id: int
    is_active: bool

    class Config:
        from_attributes = True

@app.post("/api/v1/users/", response_model=User)
async def create_user(user: UserCreate, db: Session = Depends(get_db)):
    db_user = crud.create_user(db, user)
    return db_user
```

SQLAlchemy ORM

```python
from sqlalchemy import Column, Integer, String, Boolean, DateTime
from sqlalchemy.orm import declarative_base

Base = declarative_base()

class User(Base):
    __tablename__ = "user"

    id = Column(Integer, primary_key=True)
    email = Column(String, unique=True, index=True)
    full_name = Column(String)
    hashed_password = Column(String)
    is_active = Column(Boolean, default=True)
    created_at = Column(DateTime, default=datetime.utcnow)
```

JWT □□

```python
from fastapi.security import HTTPBearer, HTTPAuthCredentials
from jose import JWTError, jwt
from datetime import timedelta

security = HTTPBearer()

def create_access_token(data: dict, expires_delta: timedelta = None):
    to_encode = data.copy()
    if expires_delta:
        expire = datetime.utcnow() + expires_delta
    else:
        expire = datetime.utcnow() + timedelta(hours=24)

    to_encode.update({"exp": expire})
    encoded_jwt = jwt.encode(to_encode, SECRET_KEY, algorithm=ALGORITHM)
    return encoded_jwt

async def get_current_user(credentials: HTTPAuthCredentials = Depends(security)):
    token = credentials.credentials
    try:
        payload = jwt.decode(token, SECRET_KEY, algorithms=[ALGORITHM])
        username: str = payload.get("sub")
        if username is None:
            raise HTTPException(status_code=401)
    except JWTError:
        raise HTTPException(status_code=401)

    user = crud.get_user_by_email(db, username)
    return user
```

React + Axios □□□□□

```typescript
// src/api/users.ts
import axios from 'axios';

const API_URL = process.env.REACT_APP_API_URL || 'http://localhost:8000/api/v1';

interface User {
  id: number;
  email: string;
  full_name: string;
}

export const createUser = async (email: string, password: string): Promise<User> => {
  const response = await axios.post(`${API_URL}/users/`, {
    email,
    password,
    full_name: '',
  });
  return response.data;
};

export const getUser = async (token: string): Promise<User> => {
  const response = await axios.get(`${API_URL}/users/me`, {
    headers: {
      Authorization: `Bearer ${token}`,
    },
  });
  return response.data;
};
```

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□□□□ **1: SaaS □□□□□□□ □□**
- □□ □□□□ □□□□ □□
- Stripe □□ □□
- □□□ □□ □ □□ □□ □□

□□□□ **2: □□□□□□□□ □□**
- FastAPI □□□□ □□ □□□□ □□
- □ □□□□ □□□ □□
- API Gateway □□

□□□□ **3: □□ □□□**
- Redis □□ □□ □□
- □□□□□□ □□ □ □□□
- CDN□□ □□ □□ □□

Claude Code □□ □

```
/plan-eng-review → □□
- □□□
- □□

/qa → □□□
- FastAPI □□
- React □□
```

3-1-3. Redoc: API □□□ □□

GitHub Stars: 24K+
□□□□ □□: OpenAPI □□□ □□□□ □□□ □□□□ □□
□□: API □□ □□□, □□□ □□ □□

□□

Redoc□ OpenAPI 3.0 □□□ □□□□ □□□□□□ API □□□ □□ □□□□□. Swagger UI□ □□ Try It Out□ □□□, □□ □□ □□ □□□ □□□□ UI□ □□□□□.

□□ □□

```
# npm□□ □□
npm install redoc redoc-cli

# CLI□ □□ HTML □□
npx redoc-cli build openapi.yaml -o index.html

# □□ React □□□□□ □□
npm install redoc react
```

□□ □□ □□

OpenAPI □□ □□ (openapi.yaml)

RealWorld □□□ □□ □□□ □□□□ □□□□□. □□ YAML□ □□□ `api/` □□□□ □□ □□□ □ □□□□.

- `info`: title, version, description
- `servers`: Production/Staging URL
- `paths`: `/api/articles` (GET/POST), `/api/articles/{slug}`, `/api/users/login` □
- `components.schemas`: Article, Profile, ArticleCreate (□□ □□: title/description/body)
- `components.securitySchemes`: BearerAuth (JWT)

React□□ Redoc □□□

```
import React from 'react';
import { ReDocStandalone } from 'redoc';

export const ApiDocumentation = () => {
  return (
    <ReDocStandalone
      specUrl="https://api.realworld.io/openapi.yaml"
      options={{
        scrollYOffset: 50,
        hideDownloadButton: false,
        hideHostname: false,
        hideLoading: false,
        suppressWarnings: false,
        expandResponses: '200,201',
      }}
    />
  );
};
```

FastAPI□□ □□ OpenAPI □□

```
from fastapi import FastAPI
from fastapi.openapi.utils import get_openapi

app = FastAPI(
    title="RealWorld API",
    description="Medium Clone API",
    version="1.0.0",
)

def custom_openapi():
    if app.openapi_schema:
        return app.openapi_schema

    openapi_schema = get_openapi(
        title="RealWorld API",
        version="1.0.0",
        description="Medium Clone API",
        routes=app.routes,
    )

    openapi_schema["info"]["x-logo"] = {
        "url": "https://fastapi.tiangolo.com/img/logo-margin/logo-teal.png"
    }

    app.openapi_schema = openapi_schema
    return app.openapi_schema

app.openapi = custom_openapi

# http://localhost:8000/redoc □□ □□ □□ □□□
```

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□□□□ **1:** □□ **API** □□□
- FastAPI/Spring Boot □□□□ □□ □□
- CI/CD □□□□□□ □□
- □□ □ □□ □□ □□ □□

□□□□ **2:** □□□ □□ □□
- Redoc + □□ □□□ □□□
- □□ □□ □□□
- □□□□ SDK □□

□□□□ **3:** □□ □□
- □□ API □□ □□ □□ □□
- □□□□□□ □□□ □□

Claude Code □□ □

```
/think → OpenAPI □□
- □□□
- ✍□□□

/document-release → □□□
- OpenAPI YAML □□
- □HTML □
```

3-2. □□□□□ & □□□□ □□

3-2-1. First Contributions

GitHub Stars: 46K+
□□□□ □□: □□□□ □ □□ □□□ - PR □□□ □ □□□
□□: □/GitHub □□ □□, □□□□ □□ □□

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first-contributions/first-contributions□ □□□□ □ PR□ □□ □□ □ □□□ □□□□ □□□□□□□. 15□ □□ □□□ □□□□□ □, □□□ PR □□□ □□□□ □□ □□□ □□ □ □□□□.

□□ □□

```
# 1. □□ Fork (GitHub □□□□□□)
# https://github.com/firstcontributions/first-contributions

# 2. □□□ □□
git clone https://github.com/<your-username>/first-contributions.git
cd first-contributions

# 3. upstream □□□ □□
git remote add upstream https://github.com/firstcontributions/first-contributions.git
```

□□ □□□□□

Step 1: □□□ □□

```
git checkout -b add-<your-name>
```

Step 2: Contributors □□□ □□ □□

`CONTRIBUTORS.md` □□□ Contributors □□□ `- [Your Name](https://github.com/your-username)` □□□□ □ □ □□□□□.

Step 3: □□ □□

```
git add CONTRIBUTORS.md
git commit -m "Add <your-name> to Contributors"
```

Step 4: GitHub□ □□ □ PR □□

```
git push origin add-<your-name>
```

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□□□□ **1:** □ □□□□ □□
- □ □□□□ □ PR □□
- GitHub □□□□ □□□
- □□□□ □□ □□

□□□□ **2:** □□□□ □□ □□ □□□
- □ □ □□□□□ □□
- □□ □□□ → □□ → PR
- □□□□□ □□□

Claude Code □□ □

```
/think → Git □□
- Fork vs Clone □
- Upstream sync □

/document-release → PR □□
- □□□
- □□□
```

3-2-2. Awesome for Beginners

GitHub Stars: 70K+
□□□□ □□: □□□ □□ □□□□□ □□ □□
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□□

MunGell/awesome-for-beginners□ "good first issue" □□□ □□ □□□□□ □□□□□ □ □□ □□□□□□□. □ □□□□□ □□ □, □□, □□□ □□ □□□□□ □□ □□□ □ □□ □□□□□□ □□ □ □□□□□.

□□□□□ □□ □□□ (Python □□□□ □□)

- **First Contributions** — □□□ ⭐, □□ □□, Git/GitHub □□ □□
- **Up For Grabs** — □□□ ⭐⭐, JavaScript/Python/C#, □□□ □□□ □□ □□□
- **Exercism** — □□□ ⭐⭐⭐, Python/JavaScript/Ruby □, □□ □□ + □□ □□
- **PySimpleGUI** — □□□ ⭐⭐, Python, GUI □□□□□ □□

□□ □□

1□□: □□ □□ □□

```
# GitHub □□
site:github.com label:"good first issue" language:python

# □□ awesome-for-beginners□□ □□ □□ □□
```

2□□: □□ □□ □□□□□

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3□□: □□ □□

```
# □: □□□ □□ □□
# Before
def calculate_total(items):
    total = 0
    for item in items:
        total = total + item.price  # □□: □□ □□□
    return total

# After
def calculate_total(items):
    total = 0
    for item in items:
        total = total + (item.price * (1 - item.discount))
    return total
```

4□□: PR □□

PR □□□□ □□ □□□ □□□□ □□□ □□□□□.

- □□: □□ □□(#123)□ □□□□□
- □□ □□: □□ □□ 3-5□ bullet
- □□□ □□: `pytest -v` □ □□ □□ □□
- □□□□□: □□□/□□/□□ □□□ □□

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□□□□ **2: □□ □□ □□**
- Python □□□ □□ □□□□□ □□
- □□ □□ □□ □□ □□
- □□ □□ □□□ □□□ □□

Claude Code □□ □

```
/think → □□
- □□
- □□□

/review → PR □□
- □□□
- □□□

/qa → □□□
- □□□
- □□□
```

3-2-3. GitHub Opensource Guide

GitHub Stars: 14K+
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□□

github/opensource.guide□ □□□□ □□□□□ □□□□ □□□□ □□□ □□□□. □□□□, □□ □□□, □□□□ □□, □□□ □□ □□ □□□ □□.

□□ □□ □□□□□

1. □□ □□: README.md, LICENSE(MIT/Apache 2.0/GPL □), .gitignore, CONTRIBUTING.md
2. □□ □□: □□□ □□ □□, □□ □□□, □□ □□, API □□
3. □□□□: CODE_OF_CONDUCT.md, Issue □□□, PR □□□, Discussions
4. □□□: GitHub Actions CI/CD, □□ □□□, □□ □□, □□□ □□□

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README.md □□□ □□ □□

- □□□□ □□□ + Shields.io □□(Stars, License, Build □)
- □ □ □□(tagline)
- □□ □□(bullet)
- □□ □□□ (bash □□ □□)
- □□ □□ (□ □□ □□ □□)
- □□ □□ (CONTRIBUTING.md □□)
- □□□□ □□

CONTRIBUTING.md □□ □□

- □□ □□ □□(clone/install/dev □□)
- □□ □□□ □□: Conventional Commits (`feat:`, `fix:`, `docs:`, `style:`, `refactor:`, `test:`, `chore:`)
- PR □□□□: Fork → □□□ □□ → □□ → PR → □□ → □□

CODE_OF_CONDUCT.md □□ □□

- □□□ □□(□□, □□□ □□□)
- □□□□ □□ □□(□□□, □□, □□)
- □□ □□ □□□

> *Claude Code □□ □*
> `> "github/opensource.guide□ □□□□ □ □□□□(□□: xxx)□ README/CONTRIBUTING/CODE_OF_CONDUCT □□□ □□□□"`

□□ □□ □□□□

□□□□ **1:** □□ □□□□ □□□□□
- □□ □□□□□ GitHub□ □□
- □□□ □□ □□ □□ □□
- □ □□□ □□□

□□□□ **2:** □ □□□□ □□
- □□□□ □□/PR □□
- □□□ □□□□ □□
- □□□□ □□ □□

Claude Code □□ □

```
/document-release → □□
- README □□
- API □□□
- □□□

/cso → □□□□
- □□□
- □□□
```

3-3. □□□ & □□□

3-3-1. Hackathon Starter

GitHub Stars: 35K+
□□□□ □□: Node.js + Express + MongoDB □□□ □□□□□□□
□□: □□□ □□ □□, Passport.js □□ □□

□□

sahat/hackathon-starter□ □ □ □□(□□□□, □□□, □□ □□□, API)□ □□□ □□□ □□□ □□□□□□. □□□□□ 2-3□□ □□ □ □ □□□□ □□□□ □□ □□ □□□ □□□ □ □□□□.

□□ □ □□

```
# □□
git clone https://github.com/sahat/hackathon-starter.git
cd hackathon-starter

# □□□□ □□
npm install

# .env □□ □□
cp .env.example .env

# MongoDB □□ (□□ □□ MongoDB Atlas)
# MONGODB_URI=mongodb://localhost/hackathon

# □□ □□ □□
npm run dev
```

□□ □□: □□

Passport □□
- `config/passport.js` □□ `LocalStrategy` □ email/password □□□ □□□□, `serializeUser` / `deserializeUser` □ □□□ □□□□□.
- □□□ □□ □□□ □□□ □□□□ □ □ □□□ □□□□ □□ □□□□□ □□ □□□ □□□□.

User □□
- `models/User.js` □□ Mongoose □□□□ email/password/profile/provider□ □□□□□.
- `pre('save')` □□□ bcrypt □□, `comparePassword` □□□□ □□□ □□.

□□□ □ □□□□
- `/signup`, `/login`, `/logout`, `/profile` (GET/PUT) □□□□□□ `routes/user.js` □ □□ □□□ □□□ □.
- □□ □□□ `ensureAuthenticated` □□□□ □□□ □□.

□□ □□□ **(Google)**
- `passport-google-oauth20` □ □□□□ `GOOGLE_ID` / `GOOGLE_SECRET` □□□□□ □□□□ □□□.
- □□ URL□ `/auth/google/callback` □□ □□, □□□ Facebook/GitHub/Twitter □□□ □□ □□ □□□□.

> *Claude Code □□ □*
> `> "hackathon-starter□ passport □□□ □ □□□□□ □□ □□□ □□□ □□ □□□□ □□□□"`

□□ □□ □□□□

□□□□ **1: 24□□ □□□**
- □□□□□ 1□□ □ □□ □ □□
- □□□ 23□□ □□ □□(AI, □□□ □□ □) □□
- □□ □□□ □□

□□□□ **2: API □□□**
- □□ □□ □□ □□□
- □□□ API □□□□□ □□
- □□ □□□□□ □□

□□□□ **3: □□□**
- Node.js + Express □□ □□
- Passport.js □□ □□ □□
- MongoDB □□□ □□ □□

Claude Code □□ □

```
/think → □□□
- □□□
- □

/plan-eng-review → □□
- API □□
- □□□

/qa → □□□
- □□□
- □□□
```

3-3-2. AutoGPT

GitHub Stars: 173K+
□□□□ □□: □□ AI □□□□ □□ - OpenAI API □□
□□: AI □□□, LLM □□□□, □□□ □□

□□

Significant-Gravitas/AutoGPT□ "AI □□□ □□ □□□ □□ □□□□ □□□ □□"□□ □□□ □□□□□. □□□□ □□□ □□ AI□ □□□ □ □□□□ □□□ □□□□ □□□□□. □□ □□□ □□□□ □□□□ □□□□□□.

□□

```
# □□
git clone https://github.com/Significant-Gravitas/AutoGPT.git
cd AutoGPT

# Python □□
python -m venv venv
source venv/bin/activate  # □□ venv\Scripts\activate (Windows)

# □□□ □□
pip install -r requirements.txt

# OpenAI API □ □□
export OPENAI_API_KEY="sk-..."

# □□
python -m autogpt
```

□□ □□□□

Agent □□
- LangChain□ `initialize_agent` □ LLM + Tools + Memory□ □□□ □□□□□ □□□□.
- Tool□ `(name, func, description)` □□□□ □□□□□. □: Calculator, Search, FileWrite.
- `agent.run(goal)` □ □□ □□□ □□□ □□□□□ □□□ □□□ □□□ □□□□□.

Task Planning
- `PromptTemplate` + `LLMChain` □□ "□□ → □□□ □□" □□□ □□□□□.
- □□ □□□□ □ □□ □□□ □□□ □□□□ □□ steps □□□□ □□□ □□□□□.

Memory System
- `ConversationBufferMemory` □ □□ □□□ □□□ □□, `ConversationSummaryMemory` □ □□□□ □□□□□.
- AutoGPT□ □ □□□□ □□ □□ □ □□□□□ □□□ □□□□□.

> *Claude Code □□ □*
> `> "AutoGPT□ Agent □□□□ □□ □ □□□□(□□ □□ → □□□□ □□□ □□)□ □□ □□ □□□□ □□□"`

□□ □□ □□□□

□□□□ 1: AI □□□ □□
- AutoGPT □□ □□□ □□ □□
- □□ □□□(□: □□□ □□, □□□ □□) □□□
- □□□ □□□ □□ □□

□□□□ 2: □□ □□□
- □□ □□□ □□ □□

- □□□ □□ □□□
- □□□ □□ □□□□□ □□

□□□□ **3:** □□/□□□□□
- LLM □□ □□□□ □□ □□
- AI □□ □□ □□
- AI □□ □□ □□

Claude Code □□ □

```
/think → □□
- Task planning □
- Memory system □
- Tool integration □

/review → □□
- Few-shot □□
- □□□
- Hallucination □

/qa → □□
- □□□
- □□□
- □□□
```

3-3-3. LangChain

GitHub Stars: 100K+
□□□□ □□: LLM □□□□□□ □□□□□
□□: AI □□□, RAG □□□, LLM □□ □□

□□

langchain-ai/langchain□ LLM(Large Language Model) □□ □□□□□□□ □□□ □□□□ □□□□□□□□.
Prompt □□, □□□, □□, □□ □□□ □□□□□ □□□□□.

□□

```
pip install langchain openai python-dotenv
```

□□ □□□

Simple Chain

```python
from langchain.llms import OpenAI
from langchain.chains import LLMChain
from langchain.prompts import PromptTemplate

llm = OpenAI(temperature=0)

# □□□ □□
prompt = PromptTemplate(
    input_variables=["topic"],
    template="□□□□ {topic}"
)

# □□ □□
chain = LLMChain(llm=llm, prompt=prompt)

# □□
result = chain.run("□")
print(result)
```

Document Processing with RAG

```python
from langchain.document_loaders import PDFLoader
from langchain.text_splitter import CharacterTextSplitter
from langchain.embeddings.openai import OpenAIEmbeddings
from langchain.vectorstores import FAISS
from langchain.chains import RetrievalQA

# PDF □□
loader = PDFLoader("document.pdf")
documents = loader.load()

# □□□ □□
text_splitter = CharacterTextSplitter(
    chunk_size=1000,
    chunk_overlap=200
)
docs = text_splitter.split_documents(documents)

# □□ □□□
embeddings = OpenAIEmbeddings()
db = FAISS.from_documents(docs, embeddings)

# RAG □□ □□
qa = RetrievalQA.from_chain_type(
    llm=OpenAI(),
    chain_type="stuff",
    retriever=db.as_retriever()
)

# □□ □□ □□□□
answer = qa.run("□□□?")
print(answer)
```

Agent with Tools

```python
from langchain.agents import initialize_agent, Tool
from langchain.tools import DuckDuckGoSearchRun
from langchain.llms import OpenAI

# □□ □□
search = DuckDuckGoSearchRun()

tools = [
    Tool(
        name="Search",
        func=search.run,
        description="□□□"
    ),
    Tool(
        name="Calculator",
        func=lambda x: eval(x),
        description="□□"
    )
]

# □□□□ □□□
agent = initialize_agent(
    tools,
    OpenAI(temperature=0),
    agent="zero-shot-react-description",
    verbose=True
)

# □□
result = agent.run("2024□□GDP □")
```

Memory Management

```python
from langchain.memory import ConversationBufferMemory, ConversationSummaryMemory
from langchain.chains import ConversationChain

# □□□ □□
memory = ConversationBufferMemory()

# □□□ □□
conversation = ConversationChain(
    llm=OpenAI(),
    memory=memory,
    verbose=True
)

# □□ □□
conversation.predict(input="□ □□□")
conversation.predict(input="□□□")
```

Custom Chain

```python
from langchain.schema import BaseMemory
from langchain.chains import Chain

class CustomChain(Chain):
    @property
    def input_keys(self):
        return ["user_input"]

    @property
    def output_keys(self):
        return ["response"]

    def _call(self, inputs):
        user_input = inputs["user_input"]

        # □□□ □□
        response = f"□□□ {user_input}"

        return {"response": response}

# □□
chain = CustomChain()
result = chain({"user_input": "□"})
```

□□ □□ □□□□

□□□□ **1:** □□ □□
- □□□□ □□ □□□
- □□□ □□ □□
- □□□□□ □□ □□

□□□□ **2:** □□ □□ □□□
- RAG□ □□□ □□ □□
- □□□ □□ □□
- □□ □□ □□

□□□□ **3:** □□□ □□□□
- □□ □□□□□ □□
- API □□ □□□
- □□□ □□□□□ □□□

Claude Code □□ □

```
/think → □□
- Chain □□
- Tool selection
- Memory strategy

/review → □□
- Temperature □
- Few-shot □□
- □□□

/qa → □□
- □□□
- □□□
```

□□□ □□□□ 3□ □□□□□□□.

Part 4: □□ □□ □□□□□□ □□ □□□

> □ □□□ □□□□□□ □□□□□□. □□□□ □□□□□ □□□□□□□□□□ □ □□□□□ □□□□ □□□□□ □ □□□□ □□□□□□□□□□.
> `scripts/` □□□□□□ □□ □□□□□□□□ □□ □ □□□□□□□□□□ □□ □□ □□□□□□.

□□ □□

- Python/Java/JavaScript □□ □□ □□□□□□ □□
- Word, Excel, PDF, PowerPoint □□ □□
- □□ □□□ □□□□□□ □□

4-1. Python □□ □□□□□□

4-1-1. python-docx (Word □□)

□□

```
pip install python-docx
```

□□ API

```
from docx import Document
from docx.shared import Inches
from docx.enum.text import WD_ALIGN_PARAGRAPH

doc = Document()
title = doc.add_heading('□□□
```

Excel □□ □ □□

```
from openpyxl import load_workbook
import statistics

wb = load_workbook('sales.xlsx')
ws = wb.active

# □□□ □□
data = []
for row in ws.iter_rows(min_row=2, values_only=True):
    if row[0]:  # □□ □□
        data.append(row)

# □□
sales = [row[4] for row in data]  # □□ □
print(f"□□ {sum(sales)}")
print(f"□□ {statistics.mean(sales):.0f}")
print(f"□ {statistics.median(sales):.0f}")

# □□□
high_sales = [row for row in data if row[4] > 500000]
print(f"500,000 □□ {len(high_sales)}□")
```

□□ □□□

```
from openpyxl import load_workbook
from openpyxl.worksheet.pivot import Pivot

wb = load_workbook('sales.xlsx')
ws = wb.active

# □□□ □□
pivot = Pivot()
pivot.fromSheet = ws
pivot.ref = f'A1:E{ws.max_row}'

# □□ □□□ □□
pivot_sheet = wb.create_sheet('Pivot')
pivot.location = f'{pivot_sheet.title}!A1'

wb.save('sales_pivot.xlsx')
```

4-1-3. python-pptx (PowerPoint)

□□

```
pip install python-pptx
```

□□ API

```python
from pptx import Presentation
from pptx.util import Inches
from pptx.chart.data import CategoryChartData
from pptx.enum.chart import XL_CHART_TYPE

prs = Presentation()

# □□ □□□□ (layout 0)
slide = prs.slides.add_slide(prs.slide_layouts[0])
slide.shapes.title.text = "□□"
slide.placeholders[1].text = "2024 Q1"

# □□ □□□□ (layout 5 = Blank)
slide = prs.slides.add_slide(prs.slide_layouts[5])
chart_data = CategoryChartData()
chart_data.categories = ['1□', '2□', '3□']
chart_data.add_series('□', (100, 120, 150))
slide.shapes.add_chart(XL_CHART_TYPE.COLUMN_CLUSTERED,
                       Inches(1), Inches(1), Inches(8), Inches(5), chart_data)

prs.save('presentation.pptx')
```

□□□ □: `slide_layouts[0]` (Title), `[1]` (Title+Content), `[5]` (Blank). `add_textbox`, `add_picture`, `add_chart` □ □□□□□ □□ □□ □ □□□□.

4-1-4. PyMuPDF (PDF □□)

□□

```
pip install PyMuPDF
```

PDF □□ □ □□

```python
import fitz  # PyMuPDF

# PDF □□
pdf_document = fitz.open('document.pdf')

# □□ □□
print(f"□□ {len(pdf_document)}")

# □□□ □□
for page_num in range(len(pdf_document)):
    page = pdf_document[page_num]
    text = page.get_text()
    print(f"--- □{page_num + 1} ---")
    print(text)

# □□□ □□
for page_num in range(len(pdf_document)):
    page = pdf_document[page_num]
    images = page.get_images()

    for img_index, img in enumerate(images):
        xref = img[0]
        pix = fitz.Pixmap(pdf_document, xref)

        if pix.n - pix.alpha < 4:
            pix.save(f"page{page_num}_img{img_index}.png")
        else:
            pix = fitz.Pixmap(fitz.csRGB, pix)
            pix.save(f"page{page_num}_img{img_index}.png")

pdf_document.close()
```

PDF □□

```python
import fitz

# □ PDF □□
doc = fitz.open()

# □□□ □□
page = doc.new_page()

# □□□ □□
page.insert_text((50, 50), "□□", fontsize=12)

# □□□ □□□
page.draw_rect((50, 80, 200, 120), color=(0, 0, 1), width=2)

# □□□ □□
page.insert_image((50, 150, 300, 350), filename="chart.png")

# □□
doc.save('generated.pdf')
```

4-2. Java □□ □□□□□

4-2-1. Apache POI (Excel/Word/PowerPoint)

Maven □□□

```
<dependency>
    <groupId>org.apache.poi</groupId>
    <artifactId>poi</artifactId>
    <version>5.2.3</version>
</dependency>
<dependency>
    <groupId>org.apache.poi</groupId>
    <artifactId>poi-ooxml</artifactId>
    <version>5.2.3</version>
</dependency>
```

Excel □□ API (XSSFWorkbook)

```
Workbook wb = new XSSFWorkbook();
Sheet sheet = wb.createSheet("Sales");

Row header = sheet.createRow(0);
String[] cols = {"□", "□", "□", "□"};
for (int i = 0; i < cols.length; i++) {
    header.createCell(i).setCellValue(cols[i]);
}

Row row = sheet.createRow(1);
row.createCell(0).setCellValue("2024-01-01");
row.createCell(2).setCellValue(10);

try (FileOutputStream fos = new FileOutputStream("sales.xlsx")) {
    wb.write(fos);
}
wb.close();
```

□□□ `WorkbookFactory.create(new File(...))` → `sheet.getSheetAt(0)` → `for (Row r : sheet)` □□. □□□□ `CellStyle` + `Font` + `FillPatternType.SOLID_FOREGROUND`.

Word □□ API (XWPFDocument)

```
XWPFDocument doc = new XWPFDocument();

XWPFParagraph title = doc.createParagraph();
title.setAlignment(ParagraphAlignment.CENTER);
XWPFRun run = title.createRun();
run.setText("□□");
run.setBold(true);
run.setFontSize(18);

XWPFTable table = doc.createTable(3, 2);
table.getRow(0).getCell(0).setText("□");
table.getRow(0).getCell(1).setText("□");

try (FileOutputStream fos = new FileOutputStream("report.docx")) {
    doc.write(fos);
}
doc.close();
```

□□□ □: □□ □□□□ `Paragraph → Run` 2□□□ □□. □□□□ Run□ □□□□.

4-2-2. iText (PDF □□/□□)

Maven □□□

```
<dependency>
    <groupId>com.itextpdf</groupId>
    <artifactId>itext7-core</artifactId>
    <version>7.2.3</version>
    <type>pom</type>
</dependency>
```

PDF □□

```java
import com.itextpdf.kernel.pdf.PdfDocument;
import com.itextpdf.kernel.pdf.PdfWriter;
import com.itextpdf.layout.Document;
import com.itextpdf.layout.element.Paragraph;
import com.itextpdf.layout.element.Table;
import java.io.IOException;

public class PdfGenerator {
    public static void createPdf() throws IOException {
        String dest = "output.pdf";

        PdfWriter writer = new PdfWriter(dest);
        PdfDocument pdfDoc = new PdfDocument(writer);
        Document document = new Document(pdfDoc);

        // □□
        document.add(new Paragraph("□□")
            .setBold()
            .setFontSize(18));

        // □□
        document.add(new Paragraph("□ 2024□4□6□"));

        // □□□
        Table table = new Table(2);
        table.addCell("□");
        table.addCell("□");
        table.addCell("2024-01-01");
        table.addCell("□");
        table.addCell("2024-02-01");
        table.addCell("□");

        document.add(table);
        document.close();
    }
}
```

4-3. JavaScript □□ □□□□□

4-3-1. SheetJS (xlsx)

□□

```bash
npm install xlsx
```

□□ □□□

```
import XLSX from 'xlsx';

// Excel □□ □□
const file = input.files[0];
const reader = new FileReader();

reader.onload = (e) => {
    const data = new Uint8Array(e.target.result);
    const workbook = XLSX.read(data, { type: 'array' });
    const worksheet = workbook.Sheets[workbook.SheetNames[0]];
    const json = XLSX.utils.sheet_to_json(worksheet);

    console.log(json);
};

reader.readAsArrayBuffer(file);

// Excel □□ □□
const data = [
    { □: '2024-01-01', □: 'A□', □: 10 },
    { □: '2024-01-02', □: 'B□', □: 15 },
];

const worksheet = XLSX.utils.json_to_sheet(data);
const workbook = XLSX.utils.book_new();
XLSX.utils.book_append_sheet(workbook, worksheet, "Sales");
XLSX.writeFile(workbook, "sales.xlsx");
```

4-3-2. PDFKit

□□

```
npm install pdfkit
```

PDF □□ API

```
const PDFDocument = require('pdfkit');
const fs = require('fs');

const doc = new PDFDocument();
doc.pipe(fs.createWriteStream('output.pdf'));

doc.fontSize(25).text('□□', { align: 'center' });
doc.fontSize(12).text('□ 2024-04-06');
doc.image('chart.png', { width: 400 });
doc.addPage().text('□□□');

doc.end();
```

□□□ □: `doc.text()` □ □□ □□ □□. □□□□ □□ □□(`x + i * width`)□□ □□ □□□□.

4-3-3. docx (Word □□)

□□

```
npm install docx
```

Word □□ API

```
import { Document, Packer, Paragraph, Table, TableRow, TableCell } from "docx";
import fs from "fs";

const cell = (text) => new TableCell({ children: [new Paragraph(text)] });

const doc = new Document({
  sections: [{
    children: [
      new Paragraph({ text: "□□", bold: true, size: 28 }),
      new Paragraph({ text: "□ 2024-04-06" }),
      new Table({
        rows: [
          new TableRow({ children: [cell("□"), cell("□")] }),
          new TableRow({ children: [cell("2024-01-01"), cell("□")] }),
        ],
      }),
    ],
  }],
});

Packer.toBuffer(doc).then((buf) => fs.writeFileSync("report.docx", buf));
```

□□□ □: `Document` → `sections[].children[]` □ `Paragraph` / `Table` □ □□□□□ □□ □□. `Packer.toBuffer()` □ □□□□□ □□ □□□□□.

4-4. □□ □□□ □□□□□

Claude Code□ □□ □□□□□ □□

□□ □□ □□ □□□□□

□ □□□□□ `scripts/build_guidebook.py` □ □□ □□□□□. □□□ □□□□□.

```
class DocumentPipeline:
    def __init__(self, data):
        self.data = data

    def generate_all(self):
        self._word()      # python-docx
        self._excel()     # openpyxl
        self._pptx()      # python-pptx
```

□ `_word`, `_excel`, `_pptx` □□□□ □□ □ □□ API□ □□□ □□□□□. □□□ dict □□□ □□□□ □□ □□□□ □ □□□ □□□ □□ □□.

> *Claude Code □□ □*
>
> `> "□ □□□□ □□□(dict)□ □□□ docx/xlsx/pptx 3□ □□□ □ □□ □□□□ DocumentPipeline □□□ □ □□□□"`

□□ □□: □□□□□ □□ □□

GitHub API + python-docx□ □□□ "□ Top 10 □□" □□□□□□ □□□□ □□□ □□□□□ □□□ □ □□□□.

```
import requests
from docx import Document

def generate_portfolio(username):
    repos = requests.get(f'https://api.github.com/users/{username}/repos').json()
    repos = sorted(repos, key=lambda r: r['stargazers_count'], reverse=True)[:10]

    doc = Document()
    doc.add_heading(f'{username} □, 0)
    for repo in repos:
        doc.add_heading(repo['name'], level=1)
        doc.add_paragraph(repo.get('description') or '')
        doc.add_paragraph(f"⭐ {repo['stargazers_count']} | {repo.get('language') or 'N/A'}")
        doc.add_paragraph(repo['html_url'])
    doc.save(f'{username}_portfolio.docx')

generate_portfolio('torvalds')
```

> *Claude Code □□ □*
>
> `> "□ □□□□□ □ GitHub □□□□ □□□□ Top 10 □□ □□□□□□ Word□ □□□. □ □□□□ README □ □□□ □□□□ □□□."`

Part 5: AI □□ □□ — Claude Code MCP & GSTACK

> *Claude Code□ MCP □□ □□□□ GSTACK □□□□□□ □□□□ □□□□.*
> *AI □□□ □□□ □□ □□□ □□□ □□□ □□ □ □□□□.*

□□ □□

- MCP □□ □□□ □□ □□ □□
- GSTACK 23□ □□ □□ □□
- Plan □□ & Sequential Thinking □□□
- AI □□ □□ □□

Claude Code□ □□□□ Model Context Protocol(MCP) □□□□ □□□□□ □□□□ □□□□ □□□□□.

5-1. □□ MCP □□

MCP □□

Model Context Protocol(MCP)□ Claude Code□□ □□ □□□ □□□□ □□□□ □□ □□ □□□□□□□.

```
□
Claude Code
    ↓
MCP Client
    ↓
MCP Server (GitHub, Filesystem, PostgreSQL □)
    ↓
External Services
```

GitHub MCP - □□□□□ □□

□□

```
claude mcp add github -s user -- npx -y @modelcontextprotocol/server-github
```

□□ (`~/.claude/config.json`)

```
{
  "mcpServers": {
    "github": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-github"],
      "env": {
        "GITHUB_TOKEN": "ghp_your_token_here"
      }
    }
  }
}
```

□□ □□

```
GitHub MCP□□□□
1. □□□□□
2. Issue □ □ □
3. Pull Request □
4. □□□□□
5. □□□

□□
- "kubernetes/kubernetes □□□10□□
- "tensorflow/tensorflow□PR #12345 □□
- "pytorch/pytorch□'performance' □□□□
```

Filesystem MCP - □□ □□ □□□

□□

```
claude mcp add filesystem -s local -- npx -y @modelcontextprotocol/server-filesystem /path/to/
```

□□

```
{
  "mcpServers": {
    "filesystem": {
      "command": "npx",
      "args": [
        "-y",
        "@modelcontextprotocol/server-filesystem",
        "/home/user/projects"
      ]
    }
  }
}
```

□□ □□□□

```
Filesystem MCP□□□
1. □□□□
2. □□□□
3. □□□
4. □□□
5. □□□

□□
- "src □□Python □□□
- "config.json □□□
- ".log □□□□□
```

PostgreSQL MCP - □□□□□□ □□

□□

```
claude mcp add postgres -s user -- npx -y @modelcontextprotocol/server-postgres
```

□□

```
{
  "mcpServers": {
    "postgres": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-postgres"],
      "env": {
        "DATABASE_URL": "postgresql://user:password@localhost:5432/mydb"
      }
    }
  }
}
```

□□ □□

```sql
-- MCP□ □□ Claude Code□□ □□ □□
-- 1. □□□□ □□□□ □□
SELECT * FROM information_schema.tables
WHERE table_schema = 'public';

-- 2. □□□□ □□
SELECT * FROM users
WHERE created_at > NOW() - INTERVAL '7 days'
ORDER BY created_at DESC;

-- 3. □□ □□
SELECT
    DATE(created_at) as date,
    COUNT(*) as user_count,
    AVG(age) as avg_age
FROM users
GROUP BY DATE(created_at)
ORDER BY date DESC;

-- Claude Code□ □□:
-- "□□ □ □□ □□□□ □□ □□ □□□□ □□□□□□□.
--  □□□□□ □□□□ □□□□□ □□ □□ □□ □□□□□ □□□□□□□."
```

Sequential Thinking MCP - □□□□ □□

□□

```
claude mcp add sequential-thinking -s local -- npx -y @modelcontextprotocol/server-sequential-
```

□□

```
{
  "mcpServers": {
    "sequential-thinking": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-sequential-thinking"]
    }
  }
}
```

□□ □□

```
□□□□

1. □□□
   "□□□PostgreSQL□MongoDB□□□
    □□□□?"

   → Sequential Thinking□□□□□

2. □□□
   "□12,000□□□□□
    □□□"

   → □□□□□□□

3. □□
   "□□□ □□?"

   → □□→ □□→ □□→ □□□
```

5-2. □□□ MCP □□

Playwright MCP - □□□□ □□□

□□

```
claude mcp add playwright -s user -- npx -y @modelcontextprotocol/server-playwright
```

□□ □□□□

```
□□□

1. □□
   "airbnb.com□□□□□□

2. □□
   "□□□□ username, password □□

3. □□
   "github.com□□□□□

4. □□□
   "□□□□□□
```

Claude Code □□ □□

> *"GitHub□□ 'machine learning' □□□□ □□□□, □□ 10□ □□□ □□□ Stars □□ □□ □□□□."*

Claude Code□ Playwright MCP□ □□ □□□ □□□□ □□□□□.
- □□□□ □□ □ □□□ □□
- □□□ □□ □ □□ □□
- DOM □□□□ □□□ □□
- □□□ Markdown □□ □□

> *"□ □□□ □□ □□□ □□□□□ □ □□ □□ □□□□ □□□□□ □□□□."*

Claude Code□ □□□□□ □□□ □ □□□□□ □□ □□□ □□□□□. □□□□ □□□ □□ □□□ □□□ □□□□.

Memory MCP - □□□□ □□ □□

□□

```
claude mcp add memory -s user -- npx -y @modelcontextprotocol/server-memory
```

□□

```
□□□□

1. □□□
   "□□□□□□"

   □□
   - □□
   - □□
   - □□□
   - □□

2. □□□
   "□□□□□□□□"

3. □□
   "□□□□□"
```

Docker MCP - □□□□ □□

□□

```
claude mcp add docker -s user -- npx -y @modelcontextprotocol/server-docker
```

□□ □□

```
# Docker □□□ □□
docker build -t myapp:1.0 .

# □□□□ □□
docker run -d -p 8080:8080 --name myapp myapp:1.0

# □□ □□
docker logs myapp

# □□ □□
docker stats myapp

# Claude Code□□ □□:
# "□ Node.js □□ Docker□□□□?"
# → Dockerfile □□ □□
# → docker-compose.yml □□
# → □□□ □□ □ □□
```

5-3. MCP □□ □□ □□

□□ □□□□□ (□□□ □□)

```
□ REST API □□□

1□ □□□(Filesystem MCP)
□□□ src/
□   □□□ main.py
□   □□□ models.py
□   □□□ routes.py
□□□ tests/
□   □□□ test_api.py
□□□ docker/
□   □□□ Dockerfile
□□□ requirements.txt

2□ GitHub□□□(GitHub MCP)
□□□ □□□
□□□ README □
□□□ □□

3□ PostgreSQL □□(PostgreSQL MCP)
CREATE TABLE users (
    id SERIAL PRIMARY KEY,
    username VARCHAR(50) UNIQUE,
    email VARCHAR(100) UNIQUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

4□ Docker □□(Docker MCP)
docker-compose up -d  □ PostgreSQL + API □

5□ API □□
GET    /api/users      - □□□
POST   /api/users      - □□□
GET    /api/users/:id  - □□□
PUT    /api/users/:id  - □□
DELETE /api/users/:id  - □□

6□ □□
pytest tests/ -v
```

Claude Code □□ □□ (MCP □□ □□)

> *"FastAPI + PostgreSQL□ □□□ □□ REST API □□□□□ □□□□.*
> *- Filesystem MCP□ src/, tests/, docker/ □□ □□*
> *- PostgreSQL MCP□ users □□□ □□□ □□*

> *- GitHub MCP□ □ □□ □□ □ □□ □□*
> *- Docker MCP□ docker-compose.yml □□"*

Claude Code□ □ MCP□ □□□□□□ □□□□ □□□□□□ □□□□□□. □□□□ "□□□ models.py□ □□□", "users □□□□ role □□□ □□□□□" □ □□□□□ □□ □□□ □□□ □ □□□□□.

□□□ □□□□□ □□□ □□ □□ **MCP**□ □□□ □□□□□ □□□□□. Claude□ □□□ □□□ Plan □□□ □□ □ □□□ □ □□□□□.

□□ □□ □□□□□

```
□ □□□□

1. Filesystem MCP: □□□
2. Sequential Thinking MCP: □□□
3. GitHub MCP: □□□
4. Playwright MCP: □□□
5. □□□(Markdown + PDF)
```

GSTACK□ Garry Tan□ 23□ □□ □□□, Claude Code□ □□□□□□ □□□ □ □□ □□□.

5-4. GSTACK □□ □ □□

□□ □□

```
# GSTACK □□
claude install-skill github.com/garrytan/gstack

# □□ □□
claude --version  # Claude Code □□ □□
claude skills list  # □□□ □□ □□
```

23□ GSTACK □□ □□

```
GSTACK Framework - 23 Skills

1️⃣ CEO / □□(2个)
   /office-hours
   - 创 □□□
   - □创 □□□□□□□
   - 创 □□□□□
   - 创 □产 □□

   /plan-ceo-review
   - 创 CEO □□□
   - □创 □□□□
   - 创 □□
   - 创 □□□

2️⃣ Engineer / □□(3个)
   /plan-eng-review
   - 创 □□□□□
   - □创 □□产 □产 □□

   /review
   - 创 □□□
   - □创 □产 产 产 □□
   - □□□

   /think
   - 创 □□□□
   - □创 □□□□□□
   - Sequential Thinking MCP □

3️⃣ Designer / UX (2个)
   /design-consultation
   - 创 UX/UI □□
   - 创 □产 产 □□

   /design-review
   - 创 □□□
   - 创 产 产 □□

4️⃣ QA / □□(2个)
   /qa
   - 创 □□□□
   - 创 □+ □+ □
   - 创 □□□□□

   /qa-only
   - 创 □□□
   - 创 □□□

5️⃣ Release / □(3个)
   /ship
   - 创 □□□□
   - 创 □□→ □→ □

   /land-and-deploy
   - 创 □(□) □□
   - 创 PR □→ □→ □→ □
```

```
   /canary
   - □ □□□
   - □ 5% □→ 25% → 100%

6️⃣ Security / □(1□)
   /cso
   - □ □□□(CSO) □□
   - □ □□ □□ □□

7️⃣ Documentation (2□)
   /document-release
   - □ □□□□
   - □ □ API □ □□

   /retro
   - □ □□□
   - □ □□ □ □□□

8️⃣ Planning & Analysis (7□)
   /weekly (□□)
   /daily (□□)
   /scope (□□)
   /breakdown (□□)
   /estimate (□□)
   /risk (□□)
   /deps (□□)
```

5-5. GSTACK □□□ □□ □□□

CEO □□: □□ □□

□□□□: □□□ SaaS □□□□ □□

```
# 1□□: /office-hours□ □□ □□
claude /office-hours
□
"AI □□□SaaS□□□
 □ □□(1-50□)
 □□ PDF/Word □□□, □□
 □MRR: $5,000
 □□ Python □ □□

□
✓ □□□□□
✓ GTM(Go-to-Market) □
✓ □□(MVP → v1 → Scale)
✓ □□
✓ KPI □□□

# 2□□: /plan-ceo-review□ □□ □□
claude /plan-ceo-review < sprint_plan.md

□ □□
□
✓ □□□
✓ □□□
✓ □□□
✓ □□□
✓ □□
```

Engineer □□: □□ □□

□□□□: □□□ □□□□□□□□ □□□□ □□

```
# 1□□: /plan-eng-review□ □□ □□ □□
claude /plan-eng-review
□
"□□□
 - API Gateway (Kong)
 - User Service (Node.js)
 - Document Service (Python FastAPI)
 - AI Service (Python FastAPI + PyTorch)
 - Message Queue (RabbitMQ)
 - Database (PostgreSQL + Redis)"

□
✓ □□□□
✓ □□□
✓ □□□
✓ □□□
✓ □□

# 2□□: /think□ □□ □□ □□
claude /think
□ "AI Service□□100ms □□"

□(Sequential Thinking):
1⃣ □□□
   - □□□ 50ms
   - □□ 40ms
   - I/O □ 10ms

2⃣ □□□
   - □□(GPU □)
   - □□
   - □□□

3⃣ □□
   - □1: □□□(5□)
   - □2: □□□(3□)
   - □3: □□□□(2□)

4⃣ □□
   - □ 100ms → 35ms □

# 3□□: /review□ □□ □□
claude /review < pull_request_123.diff

□
✓ □□□(Pylint, SonarQube □)
✓ □□
✓ □□□
✓ □□□
✓ □□+ □□□
```

/review □□ □□□□

> *"/review □□□□. process_documents □□□ for □□ □□□ DB□ □□□□ □□□ □□ □ □□."*

`/review` □□□ □□□ □□ □□□□ □□□ □□□□□.
- N+1 □□ □□ □□ □ □□ □□ □□
- □□ I/O□ □□□□ □□□□ □ □□ □□
- □□ □□ □ □□ □□ □□ □□
- □□□ □□□□ □□ □□ □□□

□□ □□□ "Critical 2□, Warning 5□" □□ □□□ □□ □ □□□ □□ □□□□ □□□□□. □□□□ "1□ □□□ □□□□"□□ □□□□□ □□□ □ □□□□.

QA □□: □□ □□

```
# 1□□: /qa□ □□ □□ □□
claude /qa
□ src/ □

□
✓ □□□ 8.5/10
✓ □□ 78%
✓ □□
  - 5□□(□□)
  - 12□□(□□)
✓ □□□□(□□)

# 2□□: /qa-only□ □□□□ □□
claude /qa-only
□ src/ □

□
✓ □□□ 127 passed, 3 failed
✓ □□□□(□□)
✓ □□□
✓ □□
```

Release □□: □□ □□

□□□□: v2.0 □□□ □□

```
# 1□□: /ship□ □□ □□
claude /ship
□
"v2.0 □
 - Main branch commit: abc123...
 - Release notes: RELEASE_v2.0.md
 - Deployment target: Production"

□
□□
☑ □□□ ✓
☑ □□□ ✓
☑ □□□ ✓
☑ DB □□ ✓
☑ □□□ ✓

□□
1⃣ Staging □□□
2⃣ □□□
3⃣ Production □
4⃣ □□
5⃣ □□

□□ 0□(□□)

# 2□□: /canary□ □□□ □□
claude /canary
□
"□□
 - Phase 1: 5% □(1□)
 - Phase 2: 25% □(30□)
 - Phase 3: 100% □

□□
Phase 1 (5%):
  - Error rate: 0.01% (□)
  - Latency: +2% (□)
  → Phase 2 □

Phase 2 (25%):
  - Error rate: 0.02% (□)
  - Latency: +3% (□)
  - Memory usage: +5% (□)
  → Phase 3 □

Phase 3 (100%):
  ✓ □□
  ✓ □□□
```

5-6. GSTACK + MCP □□□ □□□□□

□□□ □□□□ □□□□□□

```
📋 □ "AI □□□"
□ □□□□□□SaaS

=====================================
1□ □□(1□)
=====================================

CEO: /office-hours
├─ □□(Gmail, Outlook □□)
├─ MVP □□(□□□+ □)
└─ 6□□□

├─ /plan-ceo-review
  └─ □□□(□vs □□)

=====================================
2□ □□(2□)
=====================================

Engineer: /plan-eng-review
├─ □□□
│  ├─ Frontend: Next.js (React + TypeScript)
│  ├─ Backend: Python FastAPI
│  ├─ AI: GPT-4 API
│  ├─ Database: PostgreSQL + Redis
│  └─ Infrastructure: Docker + Kubernetes
└─ □□□

├─ /think (Sequential Thinking)
  └─ LLM □□□
    1️⃣ □□ □□□(avg 5□)
    2️⃣ □ □□□ □□
    3️⃣ □ □+ □□
    4️⃣ □□ 5□→ 2□

Designer: /design-consultation
└─ UI/UX □□

=====================================
3□ □□(6□)
=====================================

GitHub MCP + Filesystem MCP:
□□□□
├── frontend/
│   ├── components/
│   ├── pages/
│   └── styles/
├── backend/
│   ├── app/
│   │   ├── models.py
│   │   ├── routes.py
│   │   └── services.py
│   └── tests/
└── infra/
    ├── docker-compose.yml
    └── k8s/
```

```
Week 1-2: Backend API □
  Engineer: /review (□)
  □□□

Week 3-4: Frontend □
  Designer: /design-review
  UI □□

Week 5-6: AI □
  Engineer: /think
  LLM API □

=====================================
4□ □□(2□)
=====================================

QA: /qa
├─ □□□ 85%□□
├─ □□□□
└─ □□

QA: /qa-only
├─ □□□
├─ □□(1000 concurrent users)
└─ □□

Security (CSO): /cso
├─ API □□□
├─ □□□
└─ GDPR □□□

=====================================
5□ □□(1□)
=====================================

Release: /document-release
├─ □□
├─ API □□□(OpenAPI)
└─ □□

Release: /ship
├─ □□□
├─ Staging □□□
└─ Production □

Release: /canary (Progressive Rollout)
├─ Phase 1: 5% □(1□)
├─ Phase 2: 25% □(30□)
└─ Phase 3: 100% □

=====================================
6□ □□□(□□)
=====================================

□(PostgreSQL MCP + Memory MCP):
├─ □□□
├─ API □
```

```
├─ □
└─ AI □□

□/retro:
├─ □□□
├─ □□
└─ □□□
```

5-7. Plan □□ & Sequential Thinking □□□

Plan □□□?

Claude Code□ Plan □□□ □□□□□ □□□ □□□□ □□ □□□□□ □ □□ □□□□□.

```
Plan Mode □
✓ □□□□
✓ □□□□□
✓ □□□
✓ □□□□□□
```

Plan □□ □□□

```
# 1. Plan □□ □□□
claude --plan

# 2. □□□□ □□
"□□TypeScript□□"

# 3. Claude Code□ □□ □□ □□
Plan Preview:
├─ □□ tsconfig.json
├─ □□ src/**/*.ts (15□□)
├─ □□ package.json
├─ □□ src/**/*.js (15□□)
└─ □□□ 30□

# 4. □□□□ □□ □ □□
"✓ □"

# 5. □□□□ □□
□□□□□
```

Sequential Thinking □□□

Sequential Thinking□ □□□ □□□ □□□□ □□□□□ □□□□□.

□□□□: □□ □□□ □□

```
claude /think
□
"□API□□□□500ms□
 □□ Node.js + Express + PostgreSQL
 □□ 1000□□
 □ □□100ms □□

Sequential Thinking □

=====================================
Step 1: □□(□□□)
=====================================

□□□
1. Express □ 10ms
2. □□ 150ms
3. □□ 300ms
4. □□ 40ms

□□□ □□(300ms = 60%)

=====================================
Step 2: □□(□)
=====================================

DatabaseQuery □□
1. □□
   - □□□□□
   - □□ 200ms

2. □□
   - □□□□
   - □DB □

3. N+1 □□
   - □1□□+ □100□□
   - □□ 100ms

=====================================
Step 3: □□□
=====================================

□(1□):
1. □□□□
   - □□ 200ms → 50ms

2. Redis □□
   - TTL: 300□
   - □□□ 70%
   - □□ □□5ms

3. □□
   - N+1 □□(JOIN□□)
   - □□ 100ms → 20ms

□(1-2□):
1. □□□□
```

```
   - Read-only □5□
   - Load balancing□□□

2. GraphQL □
   - Over-fetching □
   - □□□

□(3□):
1. Event-driven □
   - □□□
   - □□□□

=====================================
Step 4: □□(HOW?)
=====================================

Week 1: □□
□
Mon-Tue: □□□□□
  - EXPLAIN ANALYZE□□□
  - CREATE INDEX idx_user_status ON users(status, created_at)

Wed: Redis □□
  - Lua □□□
  - □□□□

Thu-Fri: □□(N+1 □)
  - DataLoader □□
  - □□□

□
- Load □ k6□□1□RPS □
- □□□
- □□□□

□□
- □□ 500ms → 120ms (76% □)

Week 2: □□□
- □□ < 100ms
- □□ □

Week 3: □□□□

=====================================
Step 5: □□
=====================================

□□
✓ □□□< 100ms
✓ p99 □□< 200ms
✓ □□> 70%
✓ □< 0.01%

□□
- Prometheus + Grafana
- □□(CPU, Memory)
- □□□□
```

```
- □□□

=====================================
Step 6: □□
=====================================

□□
1. □□□
   - □ Cache invalidation □□

2. □□□□
   - □ □□□

3. □□(Redis)
   - □ LRU eviction □□

□□
- □□□□
- □□□□□□□
```

□□□□ □□ □□ □□□□□

```
High-Level Project Planning Workflow:

1. /daily: □□
   "□□□"
   → □□□

2. /scope: □□
   "□□□□"
   → □□

3. /breakdown: □□
   "□□□□"
   → User stories□□

4. /estimate: □□
   "□□□□"
   → Story points □

5. /think + Sequential Thinking
   "□□□□"
   → □□□

6. □□/review
   "□□□□"

7. /qa
   "□□"

8. /ship
   "□"

9. /retro
   "□□□"

This cycle repeats for continuous improvement.
```

5-8. □□ □□ □□□□ □□

□□: "AI □□ □□ □" SaaS □□

□□□□ □□: 2□ □, 3□□ □□, $2,000 □□ MRR □□

```
=====================================
Month 1: MVP □
=====================================

Week 1: CEO + Engineer □
1. /office-hours
   - □ GitHub □(□)
   - □ LLM □□□□
   - GTM: Product Hunt □

2. /plan-ceo-review
   - □□ OpenAI API □□
   - □□ □$29/$99/$299
   - KPI: 100□□□

3. /plan-eng-review
   - Backend: Python FastAPI
   - Frontend: Next.js
   - AI: GPT-4 API
   - □□ 4□

Week 2-3: □□
- Filesystem MCP: □□□□
- /think: □LLM □□
  1️⃣ □□□
  2️⃣ □□□
  3️⃣ □□□
  4️⃣ □□□

- GitHub MCP: □□□□
- /review: □□□(□)

**GSTACK □□(MVP □□)**

Claude Code□□□□□□

> "/think □ LLM □□ □□ □□□□ □□□ □□□□. □□, □□, □□□□□, □□□, □□□ 5□□ □□□ □□ □□□□□ JSON□□ □□□□□."

Sequential Thinking□□□ few-shot □ □□□□

> "□□□ □□□□□ CodeReviewer □□□□ FastAPI □□□□ □□□□. Filesystem MCP□ backend/app/services/reviewe

> "/review □□□ □□ □□□ reviewer.py □ □□□□."

> "GitHub MCP□ dev □□□□ □□□□ PR □□□ □□□□."

□**□□□□GSTACK □MCP □□*□□ □□□□□

Week 4: QA □□□
- /qa: □□80% □
- /qa-only: □□□
- /cso: □□(API □□ □□)

=====================================
Month 2: □□□□
=====================================
```

```
Week 5-6: □□□
- Memory MCP: □□□□
- /daily: □□
- /think: □□□

□□□
1. □□ 85% → 92%
2. □□ 8□→ 3□
3. □□ GitHub Actions □

Week 7-8: □□
- /document-release: □□
- □□ "LLM□□□□"
- □□□

=====================================
Month 3: □□□
=====================================

Week 9: Product Hunt □
- /ship: □□□
- /canary: □□□
  * Phase 1 (5%): □□
  * Phase 2 (25%): □□□
  * Phase 3 (100%): □□

□□
- □ 50 → 150 → 400 (□500)
- □□ 30% □
- MRR: $500 → $1,500 → $2,200

Week 10-12: □□□□
- /retro: 3□□
  * □□ LLM □□□
  * □ □□□□
  * □ □□□

□□□
- □ 1□□□
- □ GitLab, Bitbucket □
- □ □□□
- □ Kubernetes □

□12□MRR: $20,000
```

5-9. GSTACK □ □ □□□ □□□□

□□□□ □□□

```
1. □□□
   ❌ /think□□□□□
   ✅ /think□□□□□

   ❌ /review□□□
   ✅ /review□□□□

2. □□□□□□
   ❌ "□□□"
   ✅ "□□N+1 □□□
        □ 500ms, □ 100ms.
        □ Node.js + PostgreSQL + Redis"

3. □□
   /think → □□
   /plan-eng-review → □
   □
   /review → □□
   /qa → □□

4. MCP □
   /think □GitHub MCP□□□□
   Sequential Thinking □Filesystem MCP□□□
   □□PostgreSQL MCP□□□
```

Part 6: □□ □□ □ □□ □□

> □□□/□□□□□ □□□□ IT □□□□ □□ □□ □□ □□□□□□□.

□□ □□

- □□ □□ □□ □□ □□
- □□ □□□□ □□ □□
- □□□□□□/□□□□ □□□□
- □□/□□□□ □□ □□ □□

6-1. □□ □□ □□

yangshun/tech-interview-handbook (122K+ Stars)

□□: https://github.com/yangshun/tech-interview-handbook

□□□□ □□□□ □□ □□ □□□□. □□, □□, □□ □□□□□ □□ □□□□□.

📋 □□ □□□□ □□

1. □□ □□ (Coding Interview)

□□□□ □: □□ □□ □□ □□, □□ □□□□□ □□, □□□□□ LinkedIn □□, □□ □□ □□ 3-5□ □□
□□□□ □: □□ □□□□ □□, □□ □□ □□, □□ □□□□ □□, □□□□□□□ □□, □□□□□□□□ □□
□□ □: □□ □□, □□□□ □□□□ □□, □□ □□ □□, □□ □□

2. □□□□ □□□□ □□ □□□□□□ (40□)

1. □□□□□ □□□□ **(5□)**: □□□/□□□□□, Scale, QPS, Data volume
2. **API □□ (5□)**: Endpoints, Request/Response, Rate limiting
3. □□□□ □□ **(5□)**: □□□□, □□□□, Sharding □□
4. **High-level □□ (10□)**: Components, Data flow, Cache, MQ
5. □□ □□ **(10□)**: □□ □□□□□ □□, □□ □□, □□□□
6. □□□□ **(5□)**: Monitoring, Logging, Future improvements

□□ □: URL Shortener

- □□□□: □□ 500/s, □□ 100,000/s, □□□ 99.9%, Latency < 200ms
- □□ □□: 5□ □□ □ 78.8B URLs × 100B ≈ 7.88 TB
- **API**: `POST /api/shorten` (long_url → short_url), `GET /api/{code}` → 301 □□□□□
- □□□ □□: `URLMapping(id PK, short_code UNIQUE, long_url, created_at, expiration)`
- □□□□: Client → LB → API Servers → Redis Cache → MySQL, □□□ Kafka□ □□□
- □□ □□: Base62 □□□, Collision□ retry+increment, LRU □□, Consistent Hashing

---

donnemartin/system-design-primer (290K+ Stars)

□□: https://github.com/donnemartin/system-design-primer

□□ □□□ □□□ □□□ □□ □□ □□□□□□.

🏗️ □□ □□

1. Scalability (□□□)

- **Vertical Scaling**: CPU/□□□ □□□□□. □□□□□ □□ □□ □□ □□.
- **Horizontal Scaling**: □□□ □□ □ □□□□ Load Balancer□ □□. □□□ □□□ □ □□□.
- □□□ □□: Client → LB → App1/2/3 → Redis Cache → DB (Master-Slave □□)

2. □□□□□□ □□

```
SQL vs NoSQL □

SQL (□□)
□
- ACID □
- □□□
- □□
□
- □□□
- □□

□ MySQL, PostgreSQL

NoSQL (□□)
□
- □□□
- □□□
- □□
□
- ACID □□□
- □□□
- Consistency □

□ MongoDB, Cassandra, DynamoDB
```

3. □□ □□

- LRU, LFU, Write-Through, Write-Back, Cache-Aside □□ □□ □□□ □□ □□□□□.
- □□ □: □□□ □□ □ "□□/□□ □□", "□□□ □□", "□□□" □ □□□□ □□□□□.
- Python `functools.lru_cache`, `collections.OrderedDict` □ □□ □□□□ □□□ □□□□ □□□ □ □□□ □□.

---

kdn251/interviews (63K+ Stars)

□□: https://github.com/kdn251/interviews

Java □□ □□ □□□ □□□□ □ □□□□ □□.

🎯 Java □□ □□ □□

- □ □□□ LeetCode □□ □□□ Java □□□ Array/String/Tree/DP □ □□□□□□ □□□□.
- □□ □□□: □□□ □□ □□□ □ □□ □□□ □□, □□/□□ □□□□ □ □□□□□.
- □□□ □□ □□: Two Sum, Valid Palindrome, Reverse Integer, Merge Intervals, LRU Cache.

> *Claude Code □□ □*
> `> "kdn251/interviews □□□ Dynamic Programming □□□□ □□□ □□ □□ Top 5□ □□□ □□□□□□ □□ □□□□□ □□□□"`

6-2. □□/□□□ □□□ □□□□ □□□□□

B-1. □□ IT □□ □□□□□

1) QuantConnect/Lean - □□□□ □□□□ □□ (10K+ Stars)

□□ □□
- QuantConnect□ □□□ □□□□ □□□□ □□□□ □□
- C#□ Python□ □□□□ □□□□ □ □□□ □□ □□□
- □□□ □□/□□□ □□ □□□□ □□ □□□□ □□□□

□□ □□

```
# Python□ □□□ □□ □□
git clone https://github.com/QuantConnect/Lean.git
cd Lean

# C# □□ □□ (Linux/Mac)
dotnet --version  # .NET 5.0 □□

# Docker□ □□□ □□ (□□)
docker pull quantconnect/lean:latest
docker run -it quantconnect/lean:latest
```

□□ □□□ **- Python API**

```python
from AlgorithmImports import *

class BasicTradingAlgorithm(QCAlgorithm):
    def Initialize(self):
        # 1□□ □□□□ □□□□□
        self.SetStartDate(2023, 1, 1)
        self.SetEndDate(2024, 1, 1)

        # □□ □□□
        self.SetCash(100000)

        # □□□ □□ □□
        self.spy = self.AddEquity("SPY", Resolution.Daily).Symbol
        self.bond = self.AddEquity("AGG", Resolution.Daily).Symbol

    def OnData(self, data):
        # □□ □□□ □□□ 60/40 □□□□□ □□
        if not self.Portfolio.Invested:
            self.SetHoldings(self.spy, 0.6)
            self.SetHoldings(self.bond, 0.4)

        # □□□ □□ □□ (SMA □□)
        if not self.spy in data:
            return

        # □□ □□ □□ □□
        price = data[self.spy].Close
        self.Debug(f"SPY Price: {price}")
```

□□ □□ □□□□: □□□□ □□ □□ (Moving Average Crossover)

```python
from AlgorithmImports import *

class MovingAverageCrossover(QCAlgorithm):
    def Initialize(self):
        self.SetStartDate(2020, 1, 1)
        self.SetEndDate(2024, 1, 1)
        self.SetCash(100000)

        self.spy = self.AddEquity("SPY", Resolution.Daily).Symbol

        # □□ □□□□(50□)□ □□ □□□□(200□)
        self.fast_ma = self.SMA(self.spy, 50, Resolution.Daily)
        self.slow_ma = self.SMA(self.spy, 200, Resolution.Daily)

    def OnData(self, data):
        if not self.fast_ma.IsReady or not self.slow_ma.IsReady:
            return

        fast = self.fast_ma.Current.Value
        slow = self.slow_ma.Current.Value

        # Golden Cross: □□ MA□ □□ MA□ □□ □□ □□□
        if fast > slow and not self.Portfolio[self.spy].Invested:
            self.SetHoldings(self.spy, 1.0)
            self.Debug(f"Golden Cross! Buy at {data[self.spy].Close}")

        # Death Cross: □□ MA□ □□ MA□ □□□ □□ □□□
        elif fast < slow and self.Portfolio[self.spy].Invested:
            self.Liquidate(self.spy)
            self.Debug(f"Death Cross! Sell at {data[self.spy].Close}")
```

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- Lean□ □□□ □□□□ □□□ □□□□□□ □□□□ □□
- □□□□ "□□□ □□□□ □□□ □□□□ □□"□ □□ □□
- □□ □□□, □□ □□(Risk Management) □ □□ □□□ □□ □□

□□ □□ □□
- `QuantConnect/Documentation` - □□ □□
- `ranaroussi/yfinance` - Yahoo Finance □□□ □□
- `quantopian/*` - □ Quantopian □□□

---

2) OpenBB-finance/OpenBB - □□ □□ □□□ (35K+ Stars)

□□ □□
- Bloomberg □□□□ □□□□ □□□□ □□ □□ □□□
- □□, □□□□, □□ □ □□□ □□ □□ □□ □□
- □□□ □□ + API □□□ □□□□ □□

□□ □□

```
# pip□ □□□ □□
pip install openbb

# □□ □□□□□□□ □□
git clone https://github.com/OpenBB-finance/OpenBB.git
cd OpenBB
pip install -e .

# □□ □□
python -c "import openbb; print(openbb.__version__)"
```

□□ □□□ - □□ □□□ □□

```
import openbb

# □□ □□ □□ □□□
apple_data = openbb.stocks.get_historical_data("AAPL", start_date="2023-01-01")
print(apple_data.head())

# □□ □□ □□
info = openbb.stocks.get_company_info("AAPL")
print(f"Apple Market Cap: {info['marketCap']}")

# □□□□ □□
income_stmt = openbb.stocks.get_income_statement("AAPL", period="annual")
print(income_stmt)

# □□□ □□ □□
rsi = openbb.stocks.indicators.relative_strength_index("AAPL", period=14)
print(f"AAPL RSI(14): {rsi}")
```

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```python
import openbb
import pandas as pd

# □□□ 3□ □□ □□
stocks = ["AAPL", "GOOGL", "MSFT"]
comparison_data = {}

for stock in stocks:
    # □□ 1□ □□□
    data = openbb.stocks.get_historical_data(stock, start_date="2023-01-01")

    # □□□ □□
    returns = data['Adj Close'].pct_change().dropna()

    # □□ □□
    comparison_data[stock] = {
        "□□": returns.mean() * 252,  # □□□
        "□": returns.std() * (252 ** 0.5),
        "Sharpe Ratio": (returns.mean() / returns.std()) * (252 ** 0.5),
        "□□": (data['Adj Close'] / data['Adj Close'].cummax() - 1).min()
    }

# DataFrame□□ □□□□ □□
comparison_df = pd.DataFrame(comparison_data).T
print(comparison_df)

# □□ □□ □□
print("\n=== □□□===")
for stock in stocks:
    pe_ratio = openbb.stocks.get_pe_ratio(stock)
    pb_ratio = openbb.stocks.get_pb_ratio(stock)
    print(f"{stock}: P/E = {pe_ratio:.2f}, P/B = {pb_ratio:.2f}")
```

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- OpenBB API □□□ □□□□ □□ □□ □□
- □□ □□□ □□□□□ □□ □□□ □□ □□
- "□□□ □□ □□ □□□□ □□□" □□□□□ □□□□ □□ □□

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- `OpenBB-finance/OpenBBTerminal` - CLI □□
- `OpenBB-finance/docs` - □□ □□

B-2. □□□ □□□□□ □□

1) TheAlgorithms/Python - □□□□ □□ (195K+ Stars)

□□ □□

- Python□□ □□□ □□ □□□□ □□ (□□, □□, □□□□□ □)

- NCS(□□□□□□□□□) □ □□□ □□□□□ □□□ □□
- □ □□□□□ □□□□□□ □□□ □□□

□□ □□

```
git clone https://github.com/TheAlgorithms/Python.git
cd Python

# □□ □□ □□
ls -la
# output:
# sorts/          - □□ □□□□
# searches/       - □□ □□□□
# dynamic_programming/  - □□□□□
# graphs/         - □□□ □□□□
# arrays/         - □□ □□
# strings/        - □□□ □□
```

□□□ □□□□□ □□ □□ □□

```
Week 1: □& □
├─ searches/linear_search.py (□□)
├─ searches/binary_search.py (□□)
└─ sorts/bubble_sort.py, merge_sort.py, quick_sort.py

Week 2: □
├─ arrays/ (□□)
├─ linked_lists/ (□□)
└─ stacks_and_queues/ (□□)

Week 3: □
├─ dynamic_programming/fibonacci.py
├─ dynamic_programming/longest_increasing_subsequence.py
└─ dynamic_programming/knapsack.py

Week 4: □
├─ graphs/breadth_first_search.py
├─ graphs/depth_first_search.py
└─ graphs/dijkstra.py (□)

Week 5-8: □□
├─ □Level 2-3 (□□)
├─ Baekjoon □□(NCS □)
└─ LeetCode Medium □□
```

□□ □□ □□: □□ □□

```python
# TheAlgorithms/Python□ binary_search.py □□

def binary_search(arr, target):
    """
    □□□ □□□□ target□ □□ □□ □□
    □□□□□□: O(log n)
    □□□□□□: O(1)
    """
    left, right = 0, len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1  # □□□ □□ □□
        else:
            right = mid - 1  # □□ □□ □□

    return -1  # □□ □□

# □□□
sorted_array = [1, 3, 5, 7, 9, 11, 13, 15]
print(binary_search(sorted_array, 7))  # □□: 3
print(binary_search(sorted_array, 6))  # □□: -1
```

□□□ □□ □□: □□□□□ (LCM) - NCS □□

```python
from math import gcd

def lcm(a, b):
    """□□□□□ □□"""
    return (a * b) // gcd(a, b)

def lcm_multiple(numbers):
    """□□ □□ □□□□□"""
    result = numbers[0]
    for num in numbers[1:]:
        result = lcm(result, num)
    return result

# □□: □□□ □□□□□ □□
# □□: 3, 5, 7□ □□□□□□ □□□□
print(lcm_multiple([3, 5, 7]))  # □□: 105
```

□□□□□ □□: □□□□ □□

```python
def fibonacci_memo(n, memo={}):
    """□□□□□□□□ □□□ □□□□□"""
    if n in memo:
        return memo[n]
    if n <= 1:
        return n

    memo[n] = fibonacci_memo(n-1, memo) + fibonacci_memo(n-2, memo)
    return memo[n]

def fibonacci_dp(n):
    """DP □□□□ □□□ □□□□ (□ □□)"""
    if n <= 1:
        return n

    dp = [0] * (n + 1)
    dp[0], dp[1] = 0, 1

    for i in range(2, n + 1):
        dp[i] = dp[i-1] + dp[i-2]

    return dp[n]

# □□
print(f"□ {fibonacci_memo(30)}")  # 832040
print(f"DP: {fibonacci_dp(30)}")  # 832040 (□□ □□)
```

□□ □□ □□
- `neetcode-io/leetcode` - LeetCode □□ (□□ □□)
- `Baekjoon-Online-Judge/` - □□ □□ □□

---

2) neetcode-io/leetcode - LeetCode □□ □ □□ □□□ □□□ (6K+ Stars)

□□ □□
- LeetCode □□ 150□□□ □□ □□ □□
- □□□ □□□□□□□ □□□□ □□□ □□□ □□
- □ □□□□ □□ □□ □ □□/□□□□□ □□

□□ □ □□

```
git clone https://github.com/neetcode-io/leetcode.git
cd leetcode

# □□ □□
ls -la
# □□ □□:
# 1-array-hashing/
# 2-two-pointers/
# 3-sliding-window/
# 4-stack/
# 5-binary-search/
# 6-linked-list/
# 7-trees/
# 8-tries/
# 9-heap-priority-queue/
# 10-graphs/
# 11-advanced-graphs/
# 12-1-d-dp/
# 13-2-d-dp/
# 14-greedy/
# 15-intervals/
# 16-math-geometry/
```

□□□ □□□□□ □□ □□□□

```
□(□□□□):
├─ Array & Hashing (□ □)
├─ Two Pointers (□□)
├─ Sliding Window (□□)
└─ Binary Search (□□)

□(□□):
├─ Stack & Queue (□ □)
├─ Linked List (□□)
├─ Tree & Graph (□ □)
└─ Dynamic Programming (□)

□(□):
├─ Greedy (□)
├─ Math & Geometry (□ □)
└─ Trie (□)
```

Array & Hashing - Two Sum (LeetCode #1)

```python
def twoSum(nums, target):
    seen = {}
    for i, num in enumerate(nums):
        if target - num in seen:
            return [seen[target - num], i]
        seen[num] = i
    return []
```

□□ □□□□: "□□□ □ □"□ □□□□ □□□ O(n)□ □□. 3Sum/4Sum□□ □□□ □□ □□ □ □ □□□(O(n²))□ □□.

Sliding Window - Longest Substring Without Repeating Characters

```python
def lengthOfLongestSubstring(s):
    """
    □□□□ □□ □□ □ □□ □□□□ □□
    □□□□ □□□ □□ (O(n) □□)
    """
    char_index = {}
    max_length = 0
    start = 0

    for end, char in enumerate(s):
        # □□□ □□ □□□ □□ □□ □□□□
        if char in char_index and char_index[char] >= start:
            start = char_index[char] + 1

        char_index[char] = end
        max_length = max(max_length, end - start + 1)

    return max_length

# □□□
print(lengthOfLongestSubstring("abcabcbb"))  # □□: 3 ("abc")
print(lengthOfLongestSubstring("bbbbb"))      # □□: 1 ("b")
print(lengthOfLongestSubstring("pwwkew"))     # □□: 3 ("wke")
```

Binary Search - Search in Rotated Sorted Array

```python
def search(nums, target):
    """
    □□□ □□ □□□□ target □□
    □: [4,5,6,7,0,1,2] □□ 0 □□
    """
    left, right = 0, len(nums) - 1

    while left <= right:
        mid = (left + right) // 2

        if nums[mid] == target:
            return mid

        # □□ □□□ □□□□ □□ □□
        if nums[left] <= nums[mid]:
            if nums[left] <= target < nums[mid]:
                right = mid - 1
            else:
                left = mid + 1
        # □□□ □□□ □□□□ □□ □□
        else:
            if nums[mid] < target <= nums[right]:
                left = mid + 1
            else:
                right = mid - 1

    return -1

# □□□
print(search([4, 5, 6, 7, 0, 1, 2], 0))  # □□: 4
print(search([4, 5, 6, 7, 0, 1, 2], 3))  # □□: -1
```

□□ □□: □□ 1□□ □□ □□

```
Day 1-5: Array & Hashing (5□)
Day 6-10: Two Pointers (5□)
Day 11-15: Sliding Window (5□)
Day 16-20: Stack & Queue (5□)
Day 21-25: Binary Search (5□)
Day 26-35: DP (10□) - □□□
Day 36-45: Tree & Graph (10□)
Day 46-50: □□□(5□)

□50□× 1□= 2-3□□
```

□□ □□ □□

- `TheAlgorithms/Python` - □□□□ □□ (□□ □□)
- `Baekjoon-Online-Judge/` - □□ □□ □□

B-3. □□ & □□□ (□□□ □□)

1) OWASP/CheatSheetSeries - □□ □□□□ (28K+ Stars)

□□ □□
- OWASP(Open Web Application Security Project)□ □□ □□ □□□□
- □□□, □□□□□ □□□□ □□ □□ □ □□ □□□
- Web, API, Cloud □ □□□ □□ □□□ □□□□

□□ □ □□

```
git clone https://github.com/OWASP/CheatSheetSeries.git
cd CheatSheetSeries

# □□ □□□□ □□
ls -la cheatsheets/

# □: Authentication Cheat Sheet
cat cheatsheets/Authentication_Cheat_Sheet.md

# □□ □□□□□□ □□
# https://cheatsheetseries.owasp.org/
```

□□□ □□ □□ □□

```
1. Authentication Security (□□)
   ├─ Password Storage Cheat Sheet
   ├─ Session Management Cheat Sheet
   └─ Multi-Factor Authentication (MFA) □

2. Cryptography (□)
   ├─ Cryptographic Storage Cheat Sheet
   ├─ HTTPS Implementation
   └─ Key Management

3. API Security (API □)
   ├─ REST Security Cheat Sheet
   ├─ GraphQL Cheat Sheet
   └─ OAuth 2.0 Implementation

4. Data Protection (□□)
   ├─ PCI DSS □
   ├─ GDPR □
   └─ □□
```

□□ □□: □□□ □□□□ □□ **(Password Hashing)**

```python
# OWASP □□: bcrypt □□ (□□ □□ □□ □□!)

import bcrypt
from passlib.context import CryptContext

# bcrypt□ □□□ □□□□ □□
def hash_password(password: str) -> str:
    """
    □□□□□ □□□□ □□□
    OWASP: bcrypt with salt (□□ 12□□□)
    """
    # salt□ □□□□ □□□
    hashed = bcrypt.hashpw(password.encode('utf-8'), bcrypt.gensalt(rounds=12))
    return hashed.decode('utf-8')

def verify_password(password: str, hashed: str) -> bool:
    """□□□ □□□□ □□"""
    return bcrypt.checkpw(password.encode('utf-8'), hashed.encode('utf-8'))

# □□ □□
original_password = "MySecurePassword123!"
hashed_password = hash_password(original_password)

print(f"□□ {hashed_password}")
print(f"□□ {verify_password(original_password, hashed_password)}")
print(f"□□ {verify_password('WrongPassword', hashed_password)}")
```

□□ □□: SQL Injection □□ (□□□ □□□ □□)

```python
# OWASP: Parameterized Queries □□
import sqlite3

# □□□ □□ (SQL Injection □□□!)
def unsafe_query(username):
    query = f"SELECT * FROM users WHERE username = '{username}'"
    # □□: username□ ' OR '1'='1 □□ □□□ □□ □□

# □□□ □□ (OWASP □□)
def safe_query(username):
    conn = sqlite3.connect(':memory:')
    cursor = conn.cursor()

    # □□□□□□□ □□ □□ (?)
    query = "SELECT * FROM users WHERE username = ?"
    cursor.execute(query, (username,))

    return cursor.fetchall()

# □□ □□
safe_query("admin' OR '1'='1")  # □□□ (□□□□ □□□)
```

□□ □□: HTTPS □ SSL/TLS □□

```python
# Flask□ □□□ □□□ HTTPS □□
from flask import Flask
from flask_talisman import Talisman

app = Flask(__name__)

# OWASP: □□ □□ □□
Talisman(app,
    force_https=True,  # HTTP□ HTTPS□ □□ □□□□□
    strict_transport_security=True,  # HSTS □□□
    content_security_policy={
        'default-src': "'self'",
        'script-src': "'self' 'unsafe-inline'",
        'style-src': "'self' 'unsafe-inline'"
    }
)

@app.route('/')
def hello():
    return 'Secure Connection'

if __name__ == '__main__':
    # SSL/TLS □□□□ □□
    app.run(ssl_context='adhoc')  # □□ 'cert.pem', 'key.pem'
```

□□□ □□ □□ □□

```
Q1: SQL Injection□ □□
A: □SQL □□□
   □(Prepared Statements) □□

Q2: XSS(Cross-Site Scripting)□□
A: □□□□□□
   □□, □□, CSP(Content Security Policy) □

Q3: OWASP Top 10□
A: □□□□10□□□
   (Injection, Broken Authentication, Sensitive Data Exposure □)

Q4: □□□□
A: PCI DSS (□□□□□)
   - □□□□□
   - □□□
   - □□□
```

□□ □□ □□

- `OWASP/OWASP-Top-10-2021` - OWASP Top 10 □□ □□
- `ircrazierii/owasp-top-10-labs` - □□ □

---

2) trimstray/the-book-of-secret-knowledge - □□□ □□□/DevOps □□ □ □□ (160K+ Stars)

□□ □□
- Linux, □□□□, □□, DevOps □□ □□ □□ □□
- □□□ □□□ □□□□ □□ □□ □□
- □□□□ □□ □□□□ □□□, □□, □□□ □□□□

□□ □ □□

```
git clone https://github.com/trimstray/the-book-of-secret-knowledge.git
cd the-book-of-secret-knowledge

# README.md□ □□ □□□ □□
# □□□□□□□□ □□ □□
# https://github.com/trimstray/the-book-of-secret-knowledge
```

□□□ □□□ □□ □□

```
1. Linux Administration (□□)
   ├─ □□□□(chmod, chown)
   ├─ □□(ps, kill, systemd)
   ├─ □□(ip, netstat, ss)
   └─ □□(journalctl, /var/log)

2. Security (□)
   ├─ SSH □□□□
   ├─ □□(iptables, firewalld)
   ├─ □□(ACL, SELinux)
   └─ □□□

3. Network (□)
   ├─ TCP/IP □
   ├─ DNS □
   ├─ VPN □TLS/SSL
   └─ □□

4. DevOps (□)
   ├─ Docker □
   ├─ □□
   ├─ CI/CD □
   └─ □□□
```

□□ □□□: □□ □□

```
# 1. □□ □□ □□ (nmap □□)
nmap -sV localhost
# □□: □□ □□□□ □□ □□□□ □□ □□□ □□

# 2. □□□□ □□ □□ (ss □□, netstat□ deprecated)
ss -tlnp
# t: TCP, l: Listening, n: Numeric, p: Process

# 3. □□ □□□ □□ (md5sum)
md5sum /path/to/critical/file
# □□□: □□ □□□ □□□□ □□□□□ □□

# 4. □□□ □□ □□ (firewalld)
sudo firewall-cmd --list-all
sudo firewall-cmd --add-port=8080/tcp --permanent
sudo firewall-cmd --reload

# 5. □□□ □□ □□ (lynis)
sudo apt install lynis
sudo lynis audit system
```

□□ □□□: □□ □□□□

```
# 1. CPU □□□ □□□□
top -b -n 1 | head -20
# b: batch mode, n: □□ □□

# 2. □□□ □□□ □□
free -h
# □□ □□: □□□ □□□ □□□ □□

# 3. □□□ □□□ □□
df -h
du -sh /var/log/*  # □□ □□ □□ □□

# 4. □□□□ □□□ □□□□
iftop -n  # □□□ □□□□ □□□□

# 5. □□□□□ □□□ □□ □□
ps aux --sort=-%mem | head -10  # □□□ □□□ □
ps aux --sort=-%cpu | head -10  # CPU □□□ □
```

□□ □□: SSH □□ □□

```
# SSH □□ □□ □□□ (/etc/ssh/sshd_config)

# □□□ □□ □□:
# 1. □□ □□□ □□□□
sed -i 's/#PermitRootLogin yes/PermitRootLogin no/' /etc/ssh/sshd_config

# 2. □□□□ □□ □□□□ (□ □□ □□□ □□)
sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config

# 3. SSH □□ □□ (□□ 22 □□)
sed -i 's/#Port 22/Port 2222/' /etc/ssh/sshd_config

# 4. □□ □□
sshd -t

# 5. SSH □□□ □□□
sudo systemctl restart sshd

# 6. SSH □ □ □□ (□□□□□)
ssh-keygen -t rsa -b 4096 -f ~/.ssh/id_rsa
# □□: ~/.ssh/id_rsa (□□ □), ~/.ssh/id_rsa.pub (□□ □)

# 7. □□ □□ □□□ □□
ssh-copy-id -i ~/.ssh/id_rsa.pub -p 2222 user@server.com

# 8. SSH □□ (□□□□ □□ □□)
ssh -i ~/.ssh/id_rsa -p 2222 user@server.com
```

□□ □□: □□□ □□ □□□□

```
# □□□□□ □□: □□ □□ □ □□□ □□ □□

# 1. journalctl□ □□□ □□ □□
journalctl --unit=sshd -n 20  # SSH □□ 20□ □□
journalctl -u docker -n 20     # Docker □□□□ □□

# 2. □□ □□□ □□ □□
journalctl -u sshd --since "2024-01-01" --until "2024-01-02"

# 3. □□□ □□ □□□□
journalctl -f  # tail -f□ □□

# 4. □□ □□□ (□□ □□□□ □□□ □□□)
grep "Failed password" /var/log/auth.log | grep "user@ip"

# 5. □□ □□: □□□ □□ □□
journalctl -u sshd | grep "Failed\|Accepted" | tail -50
```

□□□ □□□ □□ □□

```
Q1: Linux □□(chmod)□□
A: chmod 755 = rwx r-x r-x (□□□□, □□□, □□□)
   □ □□600 (□□)

Q2: □□□□
A: □□□□□□
   □ □□(□□□)

Q3: SSH □□□□
A: □□□, Brute-force □□
   □ □□

Q4: □□□□
A: □□, □□, □□
   □ □□(audit log) □□
```

□□ □□ □□

- `OWASP/CheatSheetSeries` (□□ □□)
- `awesome-selfhosted/awesome-selfhosted` - □□ □□□ □□□

B-4. DevOps & CI/CD

1) kubernetes/kubernetes - □□□□ □□□□□□□ (113K+ Stars)

□□ □□

- □□□□ □□□ □□□□ □□□□ □□□□□□□ □□□
- □□□□□□□ □□□□□ □□ □□□
- □□□ □□□ □□□□□ □□□□ □□

□□ □ □□ □□

```
# 1. kubectl □□ (Kubernetes CLI)
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/lin
chmod +x kubectl
sudo mv kubectl /usr/local/bin/

# 2. kubectl □□ □□
kubectl version --client

# 3. □□ □□□□ Minikube □□
curl -minikube https://github.com/kubernetes/minikube/releases/download/latest/minikube-linux-
chmod +x minikube-linux-amd64
sudo mv minikube-linux-amd64 /usr/local/bin/minikube

# 4. Minikube □□
minikube start
minikube status

# 5. Kubernetes □□□□ □□
kubectl cluster-info
kubectl get nodes
```

□□ □□: Pod, Service, Deployment

```
# 1. Pod: Kubernetes□ □□ □□ □□ (Docker □□□□ □□)
# pod.yaml □□ □□
cat > pod.yaml << 'EOF'
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
  - name: nginx
    image: nginx:latest
    ports:
    - containerPort: 80
EOF

# Pod □□
kubectl apply -f pod.yaml

# Pod □□
kubectl get pods
kubectl describe pod nginx-pod

# Pod □□ □□
kubectl logs nginx-pod
```

□□ □□: Deployment □□ □□

`deployment.yaml`□ □□ □□□ □□□ □□□□.

- `apiVersion: apps/v1`, `kind: Deployment`
- `spec.replicas: 3` (Pod □□□)

- `spec.selector.matchLabels` □ `spec.template.metadata.labels` □□
- `containers[].image`, `containers[].ports.containerPort`
- `resources.requests/limits` (memory/cpu)
- `livenessProbe.httpGet` (□□□□)

□□: `kubectl apply -f deployment.yaml`, □□ □□: `kubectl get deployments/pods`, □□□□: `kubectl set image deployment/web-app web=nginx:1.21`, □□: `kubectl rollout undo deployment/web-app`.

□□ □□: **Service / ConfigMap / Secret**

- **Service** (`type: LoadBalancer`): `selector.app` □□ Pod□ □□, `port` → `targetPort` □□
- **ConfigMap**: □□ □□ □□□ (□: `DATABASE_HOST`, `LOG_LEVEL`)
- **Secret** (`type: Opaque`): □□ □□□ base64 □□□ — □□□□□ Vault/KMS □□ □□

□□□ **Kubernetes** □□ □□

- **Namespace**: □/□□□ □□ (`kind: Namespace`, `metadata.name: finance`)
- **NetworkPolicy**: □□ deny-all □ □□□□□□. `policyTypes: [Ingress, Egress]`, `podSelector: {}`
- **RBAC**: `Role`(□□ □□) + `RoleBinding`(□□□-□□ □□). □: `verbs: [get, list, watch]` □ □□

> *Claude Code □□ □*
> `> "□ Spring Boot □(8080 □□)□ K8s□ □□□□ deployment.yaml + service.yaml□ □□□, □□□ □□ NetworkPolicy□ RBAC□ □□ □□□□"`

□□ □□ □□
- `kubernetes/kubernetes` - □□ □□□
- `kelseyhightower/kubernetes-the-hard-way` - Kubernetes □□ □□

---

2) actions/runner - GitHub Actions □□ □□ (5K+ Stars)

□□ □□
- GitHub Actions□ □□ □□ □□ (runner)
- CI/CD □□□□□ □□□ □□
- □□□□□ □□□ □□□ □ □□□ □□

GitHub Actions □□ □□

```
# GitHub □□□□□ .github/workflows □□□□ □□
mkdir -p .github/workflows

# □□□□□ □□ □□
cat > .github/workflows/ci.yml << 'EOF'
name: CI Pipeline

on:
  push:
    branches: [main, develop]
  pull_request:
    branches: [main]

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
    - name: Checkout code
      uses: actions/checkout@v3

    - name: Set up Python
      uses: actions/setup-python@v4
      with:
        python-version: '3.9'

    - name: Install dependencies
      run: |
        python -m pip install --upgrade pip
        pip install -r requirements.txt

    - name: Run tests
      run: |
        pytest tests/ -v --cov=src

    - name: Security scan
      run: |
        bandit -r src/ -f json -o bandit-report.json

    - name: Upload coverage
      uses: codecov/codecov-action@v3
      with:
        file: ./coverage.xml
EOF
```

□□ □□: Python □□□□ CI/CD □□□□□

Python □□ □□□□ □□□□□□ 3□ job□□ □□□□□.

- **test job**: `actions/checkout` → `setup-python` (matrix□ 3.8-3.11) → `pip install` → `flake8`, `mypy`, `pytest --cov` → `codecov-action` □□ □□□□ □□□
- **security job**: `bandit`, `semgrep --config=p/security-audit`, `safety check`
- **deploy job**: `needs: [test, security]` □ `if: github.ref == 'refs/heads/main'` □□□ □□ □□□□ □□

□□□ □□ CI/CD □□

□□□□ □ □□ □□□□□□□ □□ □□□ □□□ □□□□□.

- □□ □□ □□□: `pylint --fail-under=9.0`, `black --check`, `isort --check-only`
- □□□ □□□: `pip-audit`, `safety check`
- □□□/□□□□□□ □□ □□□□ (`verify-encryption.sh`, `check-compliance.sh`)
- □□ □□□: `locust --headless -u 100 -r 10 --run-time 60s`
- SBOM □□: `cyclonedx-bom -o sbom.xml`
- approval-gate: PR □□ 2□ □□ □□ (`actions/github-script` □ `pulls.listReviews` □□)

> *Claude Code □□ □*
> `> "□□□ CI/CD □□□□(□□ □□/□□/SBOM/2□ □□)□ □□ □□□□ GitHub Actions workflow□ □ Python □□□□□ □□ □□□□"`

GitHub Actions □□ □□: □□ □□ □□

```yaml
# .github/workflows/release.yml
name: Automated Release

on:
  push:
    tags:
      - 'v*'

jobs:
  release:
    runs-on: ubuntu-latest

    steps:
    - uses: actions/checkout@v3

    - name: Create Release Notes
      run: |
        # □□□□□□
        git log $(git describe --tags --abbrev=0)..HEAD --oneline > RELEASE_NOTES.md

    - name: Create GitHub Release
      uses: actions/create-release@v1
      env:
        GITHUB_TOKEN: ${{ secrets.GITHUB_TOKEN }}
      with:
        tag_name: ${{ github.ref }}
        release_name: Release ${{ github.ref }}
        body_path: RELEASE_NOTES.md
```

□□ □□: GitHub Actions □□□

```
Step 1: □□□(1□)
├─ Checkout, Setup Python
├─ Install dependencies
└─ Run tests

Step 2: CI/CD □(1□)
├─ Code quality checks
├─ Security scans
└─ Automated testing

Step 3: □□(1□)
├─ Docker build & push
├─ Kubernetes deployment
└─ Health checks

Step 4: □□(1□)
├─ Matrix builds (□□)
├─ Conditional steps
└─ Secrets □

□4□□
```

□□ □□ □□

- `actions/runner` - □□ □□□
- `awesome-actions/awesome-actions` - GitHub Actions □□□□□ □□

□□

□□ A: Claude Code Plan □□ □□ □□□

A-1. Plan □□□?

Plan □□□ Claude Code□ □□□□□ □□ □□□ □ □□□□ □□□ □□ □□□□□.

□□ □□ vs Plan □□

```
┌─ □□(Direct Mode)
│  └─ □□→ □□□
│     ✓ □
│     ✓ □□□
│
└─ Plan □(Preview Mode)
   └─ □□→ □□→ □□□
      ✓ □ □□
      ✓ □□□
```

A-2. Plan □□ □□□ □□

□□ 1: □□□ □□□

```
1. Shift + Tab □□□□□
2. □Shift + Tab + Tab (□□)
3. Claude Code □"Plan Mode" □□
```

□□ 2: □□□ □□

```
□□□□□□✓
- /plan (□/plan-mode)
- "□Plan □□"
- "□□□"

✓ "/plan □□□□"
```

□□ 3: □□ □□

```
Claude Code □□□→ Plan Mode □
```

A-3. Plan □□ vs Direct □□ □□ □□

Plan □□ □□:

```
✓ □□□(□□□□)
✓ □□□□
✓ □□□(database.yml, config.json □)
✓ □□□
✓ □□□
✓ □□□
```

Direct □□ □□:

```
✓ □□□
✓ □□
✓ □□
✓ □□□
✓ □□
✓ □□□□
```

A-4. Plan □□ □□□ □□□□□

□□: React □□□□ □□□□

Step 1: Plan □□ □□□

□□□□ `/plan React □□□`□□ □□□□, Claude□ □□□ □□ □□□ □□□□□.

- □□□□ → □□□ □□
- Hooks □□ (useState, useEffect)
- □□□□□□ □□□ □□
- PropTypes □□

Step 2: □□□□ □□

□□ □ (□□□□):

```
class UserProfile extends React.Component {
  constructor(props) {
    super(props);
    this.state = { user: null };
  }
  componentDidMount() {
    fetchUser(this.props.userId).then(user => this.setState({ user }));
  }
  render() {
    return <div>{this.state.user?.name}</div>;
  }
}
```

□□ □ (□□□ + Hooks):

```
function UserProfile({ userId }) {
  const [user, setUser] = useState(null);
  useEffect(() => {
    fetchUser(userId).then(setUser);
  }, [userId]);
  return <div>{user?.name}</div>;
}
```

Step 3: □□ □□ □□

□□ □□: `Apply Changes` / `Edit Plan` / `Discard`. "Apply Changes"□ □□□□□.

Step 4: □□ □□

- □□□□ □□□□ □□
- □□: `src/components/UserProfile.jsx`
- □□ □□: 8 → 13 (5□ □□)

A-5. Plan □□ (Ctrl+G)

□□□ □□□ □ □□□ □□□ □ □□□□.

```
Plan □□
┌──────────────────────────────────────
│ 📋 PLAN: □□
│ ├─ 1. getUser() □□
│ ├─ 2. formatUserData() □□
│ └─ 3. □□□
│
│ [🔧 Ctrl+G□□] [ ✓ □] [ ✗ □]
└──────────────────────────────────────

□□
┌──────────────────────────────────────
│ 1. getUser() □□✓
│ 2. formatUserData() □□✓
│ 3. □□□✓
│ 4. (□□) JSDoc □□
│
│ [□]
└──────────────────────────────────────
```

A-6. Plan □□ □□□ □□□□

□□ Plan □□ □□:

```
# ✓ □□ □: □□□□ □□
"Plan □□□
1. User □validate_email() □unittest□□
2. □□ □□ □□ None □
3. □□ tests/test_user.py□□

# ✗ □□ □: □□ □□□
"□□
```

Plan □□ □□ □□□□□:

```
□□□□□
☑ □□□□
☑ □□□□(□□)
☑ □□□□
☑ □□□□
☑ □□□□□□

□□
☑ "?" □"..." □□□
☑ □□□□□
☑ □□□double-check
```

□□ B: Sequential Thinking □□ □□□

B-1. Sequential Thinking MCP□?

Sequential Thinking□ □□□ □□□ □□□ □□ □□□ □□□ □□□□ MCP(Message Collection Protocol) □□□□□.

□□:

```
□□
Q: "100□□□?"
A: 100 = 2² × 5² (□□)

Sequential Thinking:
Q: "100□□□?"
→ Step 1: 100□2□□? 50
→ Step 2: 50□2□□? 25
→ Step 3: 25□5□□? 5
→ Step 4: 5□5□□? 1
→ □ 2² × 5²

□ □□□, □□□
```

B-2. Sequential Thinking □□

□□ □□□:

```
# Claude Code CLI□□ □□
claude-code install sequential-thinking-mcp

# □□ □□ □□
npm install -g sequential-thinking-mcp

# □□ □□
claude-code mcp list | grep sequential
```

Claude Code □□ □□ (`claude.json`):

```
{
  "mcps": {
    "sequential-thinking": {
      "enabled": true,
      "timeout": 30000
    }
  }
}
```

B-3. Sequential Thinking □□ □□

□□□ □□ □□□□:

```
□ □□
  ↓
[Step 1] □□
  - □□□
  - □□□
  ↓
[Step 2] □□
  - □□□□
  - □□□
  ↓
[Step 3] □□
  - □□□
  - □□
  ↓
[Step 4] □□
  - □□□
  - □□
  ↓
□ □□
```

B-4. Sequential Thinking □□ □□

□□ □□□□:

```
✓ □□□□
  □ "N-Queen □□□□"

✓ □□
  □ "□vs □□"

✓ □□
  □ "□□□□□"

✓ □□□
  □ "□□□□□□"

✓ □□
  □ "□□□□"

✗ □□
  - □□("Python □□")
  - □□□(□□)
  - □□□□
```

B-5. Sequential Thinking □□ □□

□□ **1:** □□□ □□□□ □□

```
□□
"Sequential Thinking□□
 1000□□□target□□
 □□□□"

Sequential Thinking □
═══════════════════════════════

[THINKING Step 1: □□]
- □ □□(□1000), target □
- □ □□
- □□
  1. □□ O(n)
  2. □□ O(log n) ← □□
  3. □□ O(1) but □□

[THINKING Step 2: □□□□]
- □ O(log 1000) ≈ 10□□
- □ O(1)
- □ □vs □
  └─ □□□□

[THINKING Step 3: □□]
```python
def binary_search(arr, target):
    left, right = 0, len(arr) - 1

    while left <= right:
        mid = (left + right) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
    return -1
```
```

[THINKING Step 4: □□]
- □□□ □□: log₂(1000) = ~10□ → □□ □□□
- □□ □□□:
- target□ □□□ □□: -1 □□ ✓
- target□ □□/□□□: □□ □□ ✓

[□□]
□□ □□□ □□ (O(log n) □□□□□)

```
**□2: □□□□*
```

□□□ □□:
"Sequential Thinking□□ □□□□□?
□□ □□ □□□□□

'□□□ 2□ □□□ □□ □□□□ □□□□?'
□ □□□ □□□□ □□□□"

Sequential Thinking □□□□:
═══════════════════════════

[THINKING Step 1: □□ □□]
- □□□□: □□□ □□ □□ $1000
- □□ A: $600 □□ (□□)
- □□ B: $600 □□ (□□)
- □□:
- Race condition (□□□ □□)
- □ □ □□□□ -$200 (□□□!)

[THINKING Step 2: □□ □□]
Database □□□□ □□:

```
□ □A              □B
t1    select balance=1000 -
t2    -                   select balance=1000
t3    update balance=400  -
t4    -                   update balance=400
□ □balance = $400 (□)
```

[THINKING Step 3: □□ □□ 3□□]

□□ 1: Database Lock (□□□ □□)

```
BEGIN TRANSACTION;
SELECT balance FROM accounts WHERE id=1 FOR UPDATE;
-- □ □□ □□ □□□□□ □□ □□
UPDATE accounts SET balance = balance - 600 WHERE id=1;
COMMIT;
```

□□: □□, □□
□□: □□ □□ (□□ □□ □□)

□□ 2: MVCC (□□ □□ □□□ □□)

```
-- PostgreSQL, MySQL InnoDB
SET TRANSACTION ISOLATION LEVEL SERIALIZABLE;
BEGIN;
SELECT balance FROM accounts WHERE id=1;
UPDATE accounts SET balance = balance - 600 WHERE id=1;
COMMIT; -- □□ □□ □ □□ □□
```

□□: □□ □□□
□□: □□, □□ □□ □□

□□ 3: Optimistic Locking (□□□ □□)

```
-- version □□ □□
SELECT balance, version FROM accounts WHERE id=1;
UPDATE accounts
  SET balance = balance - 600, version = version + 1
  WHERE id=1 AND version = 1;
-- □□ □ □□□
```

[THINKING Step 4: □□□ □□ □□]
- PCI DSS: □□□□ □□□ □□
- □□□: ACID □□ □□
└─ Atomicity (□□□): □□ □□ □□ □□ □□

[□□]
□□□□□□ "Database Lock" □□ "MVCC"□
ACID □□ □□. □□□ □□□.

```
#### B-6. Sequential Thinking□GSTACK □

**GSTACK □+ Sequential Thinking:**
```

□□□□ □□ □□:
1. /office-hours
└─ "Sequential Thinking□□ □□□□□?
□□ □□□□□ □□□□ □□□"

1. Sequential Thinking□□ □□:
   ├─ □□□□□□□ (□□□ □□)
   ├─ □□□□ (□□□ □□)
   └─ □□□□□ (□□□)

2. /plan-ceo-review
   └─ □ □□ □□□ □□□□ □□□□ □□

3. □□ □□
   └─ □□ □□ □□□□ □□ □□

```
---
## □C: □□□

#### C-1. □→ □□(12□□)

**□□**
- □□□□
- □□□□□(□□□)
- □20-30□□□

**Month 1-2: □□(4□ 80□)**
```

□□ □□: HTML/CSS/JavaScript □□ □□

Week 1-2: HTML & CSS
├─ freeCodeCamp "Responsive Web Design" (300□□ → 40□□ □□)
├─ HTML5 □□□ □□
├─ CSS Grid & Flexbox
└─ □□□□: □□□□□ □□□□ (□□□)

Week 3-4: JavaScript □□
├─ □□, □□□, □□□
├─ □□□, □□□, □□
├─ DOM □□
└─ □□□□: □□□, Todo □□□

□□ □□:
- freeCodeCamp JavaScript (4□□ □□)
- MDN Web Docs (□□□□)
- Codepen (□□ □□)

```
**Month 3-4: React □(4□ 80□)**
```

□□ □□: React □□□□ □□ □□

Week 1-2: React □□
├─ JSX □□
├─ □□□□ (□□□)
├─ Props□ State
├─ Hooks (useState, useEffect, useContext)
└─ □□□□: □□ □, □□ □□

Week 3-4: React □□
├─ □□□ Hooks
├─ □□ □□□ (React.memo, useMemo)
├─ □□□ (React Router)
└─ □□□□: SNS □□, □□□

□□ □□:
- React Official Tutorial
- Dave Ceddia "Pure React"
- Scrimba React Course
- □□□□: GitHub API□ □□□ □□□ □□

```
**Month 5-6: Spring Boot □(4□ 80□)**
```

□□ □□: REST API □□ □ □□□□□□ □□

Week 1-2: Spring Boot □□
├─ □□□□ □□ (MVC □□)
├─ Controller, Service, Repository
├─ JPA/Hibernate ORM
├─ □□□□□□ □□ (MySQL)
└─ □□□□: □□□ CRUD API

Week 3-4: Spring Boot □□
├─ Spring Security (□□/□□)
├─ JWT □□ □□
├─ □□ □□
├─ □□ □ □□□□
└─ □□□□: □□□ □□ □□□

□□ □□:
- Spring Boot Official Guide
- Baeldung (□□□□ □□□)
- Java Spring Framework □□ (□□□/Udemy)
- □□□□: □□□ API (CRUD + □□ □□)

```
**Month 7-8: □& API (4□ 80□)**
```

□□ □□: □□□□□□ □□ □ □□ API □□

Week 1-2: □□□□□□
├─ SQL □□ (SELECT, INSERT, UPDATE, DELETE)
├─ JOIN, GROUP BY, □□□□
├─ □□□ □ □□□
├─ □□□ (1NF, 2NF, 3NF)
└─ □□: Leetcode Database □□ (Medium Level)

Week 3-4: □□ API
├─ □□□□□□
├─ □□□ □ □□
├─ □□ (Redis)

├─ API □□□ (Swagger/OpenAPI)
└─ □□□□: □□□□□ API (□□ □□□, □□ □□)

□□ □□:
- SQL Tutorial (W3Schools)
- Database Design Basics (Stanford)
- Redis □□ □□
- □□□□: □□ □□□ API (□□□ □□□)

```
**Month 9-10: □□& □(4□, 80□)**
```

□□ □□: □□□ □□□□□ □□□□ □□

Week 1: □□□□ □□ & □□
├─ □□□□ □□ (□□□□ □□ □□ □□)
├─ API □□
├─ □□□□□□ □□□ □□
└─ UI/UX □□□

Week 2-3: □□
├─ □□□ API □□
├─ □□□□□ □□
├─ □□□ □□ (Jest, JUnit)
└─ □□ □□

Week 4: □□ & □□
├─ □□ □□ (README, API □□)
├─ □□ (Heroku, AWS, Vercel)
├─ CI/CD □□ (GitHub Actions)
└─ □□□□□ □□

□□ □□□□ □□□□:
1. □□ □□ □ (□□ □□, □□/□□ □□)
2. □□□ □□ □□ □□□
3. □□□ □□ □□□
4. □□□ □□□
5. □□□ □□ □□ □□□

```
**Month 11-12: □□& □(4□)**
```

Week 1-2: □□□□ & □□□□□
├─ LeetCode Easy → Medium (50□□)
├─ □□□□ (Array, String, Tree, Graph, DP)
└─ □□ □□

Week 3: □□ □□ □□
├─ □□□ □□□ (□□□□□□□, □□, □□□□□□)
├─ □□□□□ □□□□ □□ □□
├─ □□ □□ □□
└─ □□ □□ (Behavioral Interview)

Week 4: □□ □□
├─ □□□ □□
├─ □□□□□ □□
├─ □□□□ □□ (□□ 3-5□)
└─ □□ □□□□

□□ □□:
- Leetcode + □□□□ □□
- "□□□ □□□ □□" □
- Cracking the Coding Interview

```
---

### C-2. □□□□(6□□□)

**□□**
- □□□(□□□□)
- □□□

**Month 1: Java & Spring Boot □(4□)**
```

□ □□: □□□ □□ □□ □□

Week 1: Java □□
├─ □□□□ □□□□□ (OOP)
├─ □□□ □□□□□ (List, Map, Set)
├─ □□□□□ (Thread, Synchronization)
└─ □□□ □ □□□ API

Week 2-3: Spring Boot □□
├─ Dependency Injection
├─ Spring Data JPA
├─ Spring Security (OAuth2, JWT)
├─ □□□□ □□ (ACID)
└─ □□□□: □□ □□ □□□

Week 4: □□ □□□ □□
├─ □□ □□ □□□□
├─ □□ □□□ (PCI DSS)
├─ □□□ □□
└─ □□ API (OpenBanking)

□□□□: □□ □□ □□ API
(□□, □□ □□, □□ □□)

```
**Month 2: SQL & □(4□)**
```

□ □□: □□□□ □□□□ □□ □□

Week 1-2: SQL □□□
├─ □□□□ □□ □□□□
├─ □□□□ □□ (ROW_NUMBER, RANK)
├─ □□□□ □ □□□□□ □□□□
├─ □□□□ □□
└─ □□: LeetCode Database □□ (Medium)

Week 3: □□ □□□□ □□
├─ □□/□□ □□□□ □□
├─ □□□□ □□□□ □□
├─ □□ □□ (Audit Log)
└─ □□□□ □□□□

Week 4: □□ □□
├─ □□ □□ □□ □□
├─ □□□□ □□□□
├─ □□ □□ (Redis)
└─ □□□□□: □□ □□□□ □□ □□

□□□□□: "□□□□□ □□ □□ □□" □□ □□□□
(□□ □□□□, □□□□□)

```
**Month 3: □(4□)**
```

□ □□: □□□□/□□□□ □□ □□

Week 1-2: □□□□ (60□□)
├─ □□□□□□ (DP) - 15□□
├─ □□□□ (BFS, DFS) - 15□□
├─ □□ □ □□ - 15□□
├─ □□□□ □□ - 15□□
└─ □□: Baekjoon Gold Level

Week 3: NCS □□ □□□□□□
├─ "□□□□□" □□ (□□ □□)
├─ "□□□□" □□ □□□□□
├─ □□ □□ □ □□ (60□/1□□)
└─ □□□□□: □□□□□ 5□

Week 4: □□ □□
├─ □□□□ □□□□ (□□□□□)
├─ □□ □□ (□□□□□, □□□□□)
├─ □□□ □□ □□
└─ □□ □□ □□

□□ □□□:
- LeetCode (□□, □□ □□)
- Baekjoon (□□ □□□□□, □□)
- Programmers (□□□/□□□ □□)

```
**Month 4: □& □(4□)**
```

□ □□: □□□ □□ □ □□ □□

Week 1: □□ IT □□
├─ □□□□ □□ (ISMS, PCI DSS)
├─ □□□ □□ (TLS/SSL)
├─ □□ □□ □□ (JWT, OAuth2)
└─ □□□□: □□□ □□□ □□□

Week 2: □□ □□
├─ OWASP Top 10 □□
├─ SQL Injection □□
├─ XSS, CSRF □□
└─ □□ □□ □□

Week 3: □□ □□□ □□
├─ □□ □□ (2FA)
├─ □□ □□ □□
├─ □□ □□ □□
└─ PCI DSS □□

Week 4: □□ □□
├─ "□□□□□ □□□ □□□?"
├─ "□□□ □□□□ □□ □□"
├─ "□□ □□□ □□"
└─ □□□□: □□ □□ □□□

□□□□: "□□ □□ □□ □□□"
(OTP, □□□□, □□ □□ □□)

```
**Month 5: DevOps & □(4□)**
```

□ □□: □□□ □□ □□

Week 1: Docker & Kubernetes
├─ □□□□ □□ (Docker)
├─ □□□□□□□ (Kubernetes)
├─ □□ □□□
└─ □□□□: Spring Boot □ □□□□□

Week 2: CI/CD □□□□□
├─ GitHub Actions
├─ □□ □□□
├─ □□ □□
└─ □□□□: □□□ □□ □□

Week 3: □□□□ & □□
├─ ELK □□ (Elasticsearch, Logstash, Kibana)
├─ □□ □□□□ (Prometheus)
├─ □□ □□
└─ □□□ □□ □□ □□

Week 4: □□□□ □□
├─ □□ □□□
├─ □□□□□□ □□ (Replication)
├─ □□ □□ (DR) □□
└─ □□□□: □□□□ □□□ □□

□□ □□:
- Kubernetes Official Tutorial
- GitHub Actions Documentation
- ELK Stack □□ □□

```
**Month 6: □& □(4□)**
```

Week 1-2: □□ □□□□ □□
├─ □□□□: □□ □□ □□ □□□
│ ├─ □□□ □□ □□
│ ├─ □□ □□ □□□□□
│ ├─ □□ □□ □□
│ └─ □□ □□ □ □□
├─ □□ (AWS/Azure)
├─ □□ □□□ (□□□□ < 200ms)
└─ □□□ (API □□, □□□□)

Week 3: □□ □□ □□
├─ □□□ □□□ (□□ □□□)
├─ □□ □□□ □□
├─ □□ □□

├ □□ □□ □□
└ □□ □□ 3□

Week 4: □□ □□
├ □□□/□□□ □□
├ □□□□□□ □□ □□
├ □□□ □□□□ □□
└ □□ □□ □□□□

```
**□□□□**
```

□□ □□:
☑ Java/Spring Boot (□□)
☑ SQL (□□□□□)
☑ □□□□ (Gold Level)
☑ □□ (OWASP Top 10)
☑ DevOps (Docker, K8s)

□□□□□:
☑ □□ □□ □□□□ (□□□ □□)
☑ GitHub README □□ □□
☑ □□ □□□ □□
☑ □□/□□ □□□ □□

□□ □□:
☑ □□□□ □□ □□
☑ □□□ □□□ □□
☑ □□ □□□ □□
☑ □ □□ □□ □□
☑ □□□□ □□ □□□□

□□ □□:
☑ □□ □□□ (□□, □□, □□)
☑ □□□ (□□, □□, □□□□)
☑ □□□ (□□□□□□, □□ □)
☑ □□□□ □□ □□

```
---
## □D: □□□□

#### D-1. □□□

**□□**
```

1. Velog (https://velog.io/)
   - □□□ □□□ □□□

- □□: □□ □ □□ □□
- □□□: "□□□□", "□□□□", "□□□□"

2. Tistory (https://www.tistory.com/)
   - □□ □□ □□□
   - □□ □□□: "□□□□□ □□□", "□□□□ □□"

3. Medium □□ (https://medium.com/)
   - □□ □□ □
   - □□: "Better Programming", "Towards Data Science"

4. dev.to (https://dev.to/)
   - □□□ □□□□
   - □□□□□ □□ □□

```
**□□**
```

1. GitHub □□ (https://github.com/korean-developers)
   - □□ □□□ □□□□
   - □□ □□, □□□□ □□

2. □□□□□□ (https://programmers.co.kr/)
   - □□□□□ + □□
   - □□ □□ □□□

3. □□ □□□ □□ (https://www.acmicpc.net/)
   - □□□□ □□ □□
   - □□ □□ □□

4. □□□ (https://www.inflearn.com/)
   - □□□ □□ □□
   - □□□ □□ (□□)

5. □□□ □□ (https://nomadcoders.co/)
   - □ □□ □□
   - YouTube □□ □□

```
#### D-2. □□(Online Judge)

**□□**
```

1. LeetCode (https://leetcode.com/)
   □□: □□
   □□: □□ □□, □□ □□ □□
   □□: □□□□□ Easy → Medium
   □: □□ □□ □□ ($35/□)

2. Baekjoon (https://www.acmicpc.net/)
   □□: □□
   □□: □□ □□ □□□□ □□
   □□: □□□/□□□ □□□□□ □□
   □: □□

3. Programmers (https://programmers.co.kr/learn/challenges)
   □□: □□□/□□□ □□
   □□: □□ □□□ □□
   □: □□

4. Codeforces (https://codeforces.com/)
   □□: □□ □□□□□
   □□: □□ □□□□
   □: □□

□□ □□:
Week 1-4: LeetCode Easy (50□□)
Week 5-8: LeetCode Medium (50□□)
Week 9-12: Baekjoon Silver (50□□)
Week 13-16: Baekjoon Gold (50□□)
□ 3-4□□□ 200□□

```
**□□**
```

1. HackerRank (https://www.hackerrank.com/)
   - □□ □□ □□
   - C, Java, Python □□

2. TopCoder (https://www.topcoder.com/)
   - □□□□ □□□□
   - □□ □□

3. AtCoder (https://atcoder.jp/)
   - □□ □□□□ □□
   - □□ □□□□ □□

```
### D-3. □□□

**□□**
```

1. freeCodeCamp (https://www.freecodecamp.org/)
   - □□ □□
   - □□ □□ (□□ □□)
   □□ □□:
   ├─ Responsive Web Design (5□□)
   ├─ JavaScript Algorithms (4□□)

   ├─ React (12□□)
   └─ Backend Development (12□□)

2. Khan Academy (https://www.khanacademy.org/)
   - □□□ □□ □□
   - □□ + □□ □□

3. Codecademy (https://www.codecademy.com/)
   - □□□□□ □□
   - □□ □□ □□

4. Udemy (https://www.udemy.com/)
   - □□ □□ ($10-15)
   - □□: "The Complete JavaScript Course"
   "The Complete Python Course"

□□ □□:
5. □□□ (https://www.inflearn.com/)
- □□ □□, □□□
- □□: 15,000□ (□□ □ 3,000□)
□□: "□□□□" □□□

1. □□□ □□□ (https://spartacodingclub.kr/)
   - □□□ □□□
   - □□ + □□ □□ □□

```
**□□□**
```

1. Docker (https://www.docker.com/101-tutorial/)
   - □□ □□□□ □□

2. Kubernetes (https://kubernetes.io/docs/tutorials/)
   - □□ □□, □□ □□

3. Spring Boot (https://spring.io/guides)
   - □□ □□□ □□

4. React □□ (https://react.dev/)
   - □□ React □□
   - □□□ □□□□ (2022)

```
### D-4. □□

**□**
```

1. "Cracking the Coding Interview"
   - □□: Gayle Laakmann McDowell

   - □□: □□ □□ □□
   - □□: $40 (□□□□ □□□□: 45,000□)

2. "System Design Interview"
   - □□: Alex Xu
   - □□: □□□□ □□
   - □□: $50 (□□□□ □□□□: 50,000□)

3. "Clean Code"
   - □□: Robert C. Martin
   - □□: □□ □□ □□
   - □□: $45 (□□□□: 40,000□)

4. "Effective Java"
   - □□: Joshua Bloch
   - □□: Java □□
   - □□: $55 (□□□□: 45,000□)

5. "□□□□□□□ □□□□ □□"
   - □□: □□□□
   - □□: □□ □□, SQL □□ □□
   - □□: 35,000□

```
**□& □**
```

1. "□□□□□ □□ □□ □□□□"
   - □□: □□□□
   - □□ □□□□□□□ □ □□
   - □□: 35,000□

2. "□□ □□□□ □□ □□"
   - □□: □□ □□□□
   - LeetCode □□□□ □□
   - □□: 32,000□

```
**□& DevOps:**
```

1. "□□ □□□□ □□"
   - □□: □□ □□□□□ (□□)
   - □□: Udemy $40

2. "□□□□□□□ □□ □□□□"
   - □□: Kelsey Hightower
   - □□: "Kubernetes The Hard Way"
   - □□: □□□□ □□

3. "□□□□□□□ □□□□"
   - □□: Sam Newman
   - □□: 35,000□

```
### D-5. □□□

**□□**
```

□□ □□:
1. freeCodeCamp (https://www.youtube.com/@freeCodeCamp)
- □□ □□, □□□
- □□: "100 Days of Code", "Web Development"

1. Traversy Media (https://www.youtube.com/@TraversyMedia)
   - □□□□□ □□
   - □□: "React Crash Course", "Node.js Tutorial"

2. The Net Ninja (https://www.youtube.com/@NetNinja)
   - □ □□ □□ □□
   - □□□□ □□□ □□

3. CodeWithHarry (https://www.youtube.com/@CodeWithHarry)
   - Python □□ □□
   - 100□ □□ □□

□□ □□:
5. □□□ □□ (https://www.youtube.com/@nomadcoders)
- □ □□ □□
- □□□, □□□

1. □□□□ (https://www.youtube.com/@CodingApple)
   - HTML/CSS/JS □□
   - □□ □□□ □□

2. □□ (https://www.youtube.com/@woodywood6)
   - Spring Boot, Java
   - □□□ □□ □□

3. □□□ (https://www.youtube.com/@dongbinna)
   - □□□□ □ □□
   - □□ □□□ □□□□ □□

```
**□□& □**
```

1. Google Developers (https://www.youtube.com/@GoogleDevelopers)
   - Google □□□□ □□
   - □□□ □□□ □□

2. Back to Back SWE (https://www.youtube.com/@BackToBackSWE)
   - □□ □□ □□
   - □□ □□ □□

3. Fireship (https://www.youtube.com/@Fireship)
   - □□ □□□ □□
   - 10□ □□ □□ (□□!)

```
### D-6. □□□

**□□□**
```

1. □□□ (https://www.saramin.co.kr/)
   - □□ □□
   - □□: □□□ > □□

2. □□□□ (https://www.jobkorea.co.kr/)
   - □□□ □□ □□
   - □□: □□□□

3. □□□ (https://www.work.go.kr/)
   - □□ □□ □□□
   - □□□/□□□□ □□

4. □□□□□ (https://www.gonggigeop.ai/)
   - □□□ □□ □□□
   - □□ □□

```
**□□□**
```

1. □□□ (https://www.wanted.co.kr/)
   - □□□□ □□ □□
   - □□: "□□□ □□ □□"

2. □□□□□□ □□ (https://programmers.co.kr/jobs)
   - □□□□□ + □□
   - □□ □□□

3. □□ (https://jungle.co.kr/)
   - □□ □□□□
   - □□: □□□□ + □□

4. □□□□□ (https://wellfound.com/)
   - □□ □□□□
   - □□ □□□□ □□

```
**□□□**
```

1. □□□□□□ (https://www.fss.or.kr/)
   - □□ □□ □□
   - "IT □□ □□" □□□

2. □ □□ □□□
   - □□□□ (https://www.shinhan.com/shbank/recruit)
   - □□□□ (https://www.kbbank.com/recruit)
   - □□□□ (https://www.wooribank.com/)

3. □□□ □□
   - □□ (https://toss.im/career)
   - □□ (https://about.daangn.com/jobs/)
   - □□ (https://banksalad.com/careers)

```
### D-7. □□□Tip

**□□□**
```

Tip 1: 80/20 □□
- 20%□ □□ □□□□ 80%□ □□ □□
- □: React hooks□□□ 90% □□□□ □□
- □□: □□□ □□□□ (□□□ □□)

Tip 2: □□□□ □□ □□
- □□ □□ vs □□□□ □□□ □□□
- 70% □□□□ □□□ □□
- □□: 3-4□ □□□□□ □□□□

Tip 3: □□□□ (Spaced Repetition)
- □□ □□ □□□, 1□ □, 1□□ □ □□
- □: Anki (□□□□□)
- □□: □□□□ □□ □□□

Tip 4: □□□□ (Teaching Others)
- □□ □□□ □□□□ □□
- □□□ □□□□
- □□□□ □□□□□□ □□□□□

Tip 5: □□ □□ □□
- GitHub□ □□□ □□□ □□
- □□□□ PR □□□
- □□□□ □□□□

Tip 6: □□□
- □□ 3□□ × 3□□ > □□□ 30□□
- □□ □□□ □□□ □□
- □□: □□ 1□□ + □□ 2□□

```
**□□□**
```

□□□: □□□ □□ □□
□-□□□: □□□□ □□
□□□: □□ + □□ □□
□-□□□: □□ □□□□ □□

□□ □□:
- □□ 3□□
- □□□□ 10□□
- □□□□ 7□□
- □□ □ □□□ 5□□
□ 25□□/□

```
---

---

## □E: □□□□

□□ `scripts/` □Python □□□

### □□

```bash
# □□
pip install -r scripts/requirements.txt

# □□(MD + DOCX + PDF)
python scripts/build_guidebook.py

# □□□
python scripts/build_guidebook.py --format md
python scripts/build_guidebook.py --format docx
python scripts/build_guidebook.py --format pdf
```
```

□□□□□ □□

```
src/*.md → md_merger.py → output/Guidebook.md
                                ↓
                      ┌─────────┴─────────┐
                      ↓                   ↓
              md_to_docx.py       md_to_pdf.py
                      ↓                   ↓
          output/Guidebook.docx  output/Guidebook.pdf
```

□□□ □□□□□

□□□□□	□□	**Part 4 □□**
python-docx	MD → DOCX □□	4-1
WeasyPrint	HTML → PDF □□□	4-4 (□□)
markdown	MD → HTML □□	-
Pygments	□□ □□ □□	-